Renewable Intermittency and Grid Investment Dynamics: The Role of Scarcity-Induced Battery Innovation in Vietnam

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A thesis submitted to partially fulfill the requirements for the degree of

Bachelor of Arts in Economics

Spring 2025

To all who made this possible.

Contents

Abstract

I investigate the divergence between rapid private renewable deployment and slow public grid investment which is known in engineering as the "agility gap." I develop an open-economy DSGE model with endogenous scarcity-induced technological change in battery storage and a grid reliability constraint, where variable renewable penetration creates intermittency shocks that reduce capital utilization through grid instability.

Calibrating the model to Vietnam's Power Development Plan 8, I find the following. First, the agility gap is quantitatively large and structurally asymmetric: battery investment responds quickly to reliability shortfalls but is governed almost entirely by global price shocks (100% of its variance), while grid investment responds to domestic reliability signals but accumulates slowly through the fiscal rule (82% of its variance). The two flexibility instruments are therefore driven by entirely separate forcing variables—international supply chains versus domestic fiscal decisions—which is why private storage deployment and public grid expansion cannot be treated as substitutes in policy design. Second, scarcity-induced technological change generates persistent battery improvements conditional on regulatory transmission of scarcity signals. Third, suppressing scarcity signals raises the welfare cost of intermittency by 20%, demonstrating that market mechanisms are quantitatively important complements to storage investment.

Keywords: Environmental DSGE, Scarcity-Induced Innovation, Endogenous Technological Change, Investment Inertia, Renewable Intermittency, Energy Storage.

Acknowledgements

I would like to express my deepest gratitude to my advisor, Dr. Xavier Martin G. Bautista, for his invaluable guidance, unwavering patience, and intellectual rigor throughout the development of this project. His expertise in bridging theoretical complexity with practical economic intuition has been instrumental in shaping the core of this research.

I am also grateful to the faculty of the Economics program at Fulbright University Vietnam for their support and for providing a stimulating academic environment.

To my friends and colleagues, thank you for the countless discussions and the mutual support that made this journey possible. Finally, I owe an immense debt of gratitude to my family for their constant encouragement and belief in my work.

All remaining errors are my own.

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List of Abbreviations

Abbreviations are defined at first use in the main text and listed here for quick reference.

Acronym	Meaning
BESS	Battery Energy Storage System(s)
CBAM	Carbon Border Adjustment Mechanism
CES	Constant Elasticity of Substitution
DTC	Directed Technical Change
DSGE	Dynamic Stochastic General Equilibrium
E-DSGE	Environmental DSGE
EIS	Elasticity of Intertemporal Substitution
ENS	Energy Not Served
ETS	Emissions Trading System
EVN	Vietnam Electricity (Việt Nam Điện Lực)
FEVD	Forecast Error Variance Decomposition
FiT	Feed-in Tariff
GSO	General Statistics Office of Vietnam
IAM	Integrated Assessment Model
IEA	International Energy Agency
LCOE	Levelized Cost of Energy
LNG	Liquefied Natural Gas
LOLE	Loss of Load Expectation
LOLP	Loss of Load Probability
NLDC	National Load Dispatch Centre (Vietnam)
PDP8	Vietnam Power Development Plan 8 (Decision 500/QD-TTg)
SAIDI	System Average Interruption Duration Index
SAIFI	System Average Interruption Frequency Index
SOE	Small Open Economy
TFP	Total Factor Productivity
VGB	Vietnam Government Bond
VRE	Variable Renewable Energy (primarily wind and solar photovoltaic generation whose output is non-dispatchable and weather-driven)

1 Introduction

Mitigating climate change has emerged as the defining macroeconomic challenge of the 21st century. Following the Paris Agreement, global policy has pivoted decisively toward Net Zero targets, necessitating a radical transformation of energy systems. This transition is predicated on the massive deployment of Variable Renewable Energy (VRE) sources, primarily wind and solar, whose Levelized Cost of Energy (LCOE)[1] has fallen precipitously over the last decade. However, the integration of these sources introduces a fundamental structural friction: the shift from dispatchable fossil fuel generation to non-dispatchable renewable supply. Unlike coal or gas, whose production can be adjusted to match demand, solar and wind generation are exogenously determined by nature.

As noted by Ambec and Crampes (2012), the non-storability of electricity implies that supply and demand must clear instantaneously; when VRE penetration increases without a corresponding increase in storage capacity, the marginal value of renewable energy declines. This phenomenon, empirically documented as the "cannibalization effect" (Hirth, 2013), implies that the decarbonization challenge is not merely one of installing generation capacity, but also one of managing the *reliability penalty*[2] associated with supply volatility. Empirically, Hirth (2013) documents that in German electricity markets, the market value of solar declined from approximately 100% of baseload prices in 2008 to 60% by 2016 as penetration reached 15%, demonstrating the nonlinear costs of intermittency at scale.

For emerging economies like Vietnam, this friction is acute. As it strives to sustain high growth rates while meeting ambitious decarbonization targets (PDP8), Vietnam faces a critical "energy trilemma": balancing security, affordability, and sustainability (Liu et al., 2022). The shift to renewables introduces a new source of supply-side volatility that, without adequate storage infrastructure, acts as a negative productivity shock. This capstone argues that ignoring the temporal mismatch between renewable supply and demand leads standard macroeconomic models to underestimate the welfare costs of the green transition.

[1] Levelized Cost of Energy (LCOE): an average lifetime cost per unit of electricity, typically computed as discounted total costs divided by discounted energy output.

[2] The reliability penalty is defined as the endogenous reduction in effective capital utilization (u_t) resulting from the structural mismatch between variable renewable supply and the grid's flexibility capacity. In the production function, this deficit operates equivalently to a negative Total Factor Productivity (TFP) shock.

Existing environmental DSGE (E-DSGE) frameworks typically allow for imperfect substitution between energy and other inputs through CES aggregation, following Heutel (2012). However, they overlook the non-linear costs of grid instability arising from renewable intermittency. Furthermore, most endogenous growth models in this domain, such as Acemoglu et al. (2012), focus on the inter-sectoral choice between "clean" and "dirty" production technologies, neglecting the intra-sectoral challenge of grid integration.

There remains a critical gap in the literature regarding the interaction between renewable intermittency, investment frictions, and induced innovation. Specifically, current models fail to address how a developing economy manages the *agility gap*—the divergence between the rapid deployment of private renewable assets and the slow accumulation of public transmission infrastructure. Addressing this question requires a framework that explicitly endogenizes grid reliability and allows technological progress to respond to the shadow price of flexibility through a scarcity-induced innovation mechanism.

To address these gaps, I develop a DSGE model calibrated to the structural characteristics of Vietnam. The model is built around three novel features designed to capture the mechanics of the energy transition. First, I introduce an endogenous reliability function in which capital utilization is determined by the ratio of flexibility assets to renewable-output volatility. This creates a mechanism through which aggressive renewable targets, without sufficient storage, act as a drag on total factor productivity (TFP). Second, I model scarcity-induced technological change in battery storage, where the rate of technological progress responds to the gap between target and actual grid reliability. This mechanism, distinct from the Directed Technical Change framework of Acemoglu (2002), captures induced innovation without requiring explicit R&D sectors or resource allocation decisions. Third, I incorporate public-investment inertia to distinguish between private battery capital, which is agile and responds to market signals, and public grid capital, which is subject to construction lags and follows institutional rules. This distinction allows for the analysis of structural bottlenecks in which public infrastructure fails to keep pace with private energy deployment.

This study contributes to the Environmental DSGE literature in three distinct dimensions. The first contribution is the joint modeling of an explicit energy sector with endogenous grid reliability. Unlike standard E-DSGE models that include energy as a production input without specifying its source, I model renewable and fossil generation explicitly, allowing intermittency shocks to propagate through both the energy market and the reliability constraint. This dual-channel mechanism quantifies the full macroeconomic cost of renewable variability.

The second contribution is the integration of scarcity-induced technological change with investment frictions to identify the *agility gap*. While previous literature focuses on frontier innovation races between clean and dirty technologies, I model innovation as a scarcity-signal-driven process where persistent reliability gaps direct technological progress toward battery storage. Crucially, I distinguish between private storage (batteries), which responds to price signals, and public infrastructure (grid), which is subject to planning delays. This novel specification captures the structural lags observed in developing economies like Vietnam.

The third contribution provides a "second-best" policy analysis for regulated markets. I demonstrate that the efficacy of scarcity-induced technological progress is strictly conditional on the transmission of scarcity signals. By simulating the model under dampened price pass-through (capturing regulated tariffs or incomplete remuneration for ancillary services), I identify a result of *signal attenuation*, showing that scarcity-induced innovation cannot substitute for the allocative efficiency of market-clearing prices.

The remainder of this capstone is organized as follows. Section 2 reviews the relevant literature. Section 3 outlines the model environment. Section 4 details the calibration strategy for Vietnam. Section 5 presents the model mechanics, log-linearization, and transmission channels. Section 6 analyzes the baseline quantitative results, including impulse responses and variance decomposition. Section 7 presents the welfare analysis, counterfactual experiments, and parameter sensitivity. Section 8 concludes with a discussion of policy implications, limitations, and directions for future research. Detailed solution methods are provided in the Appendix.

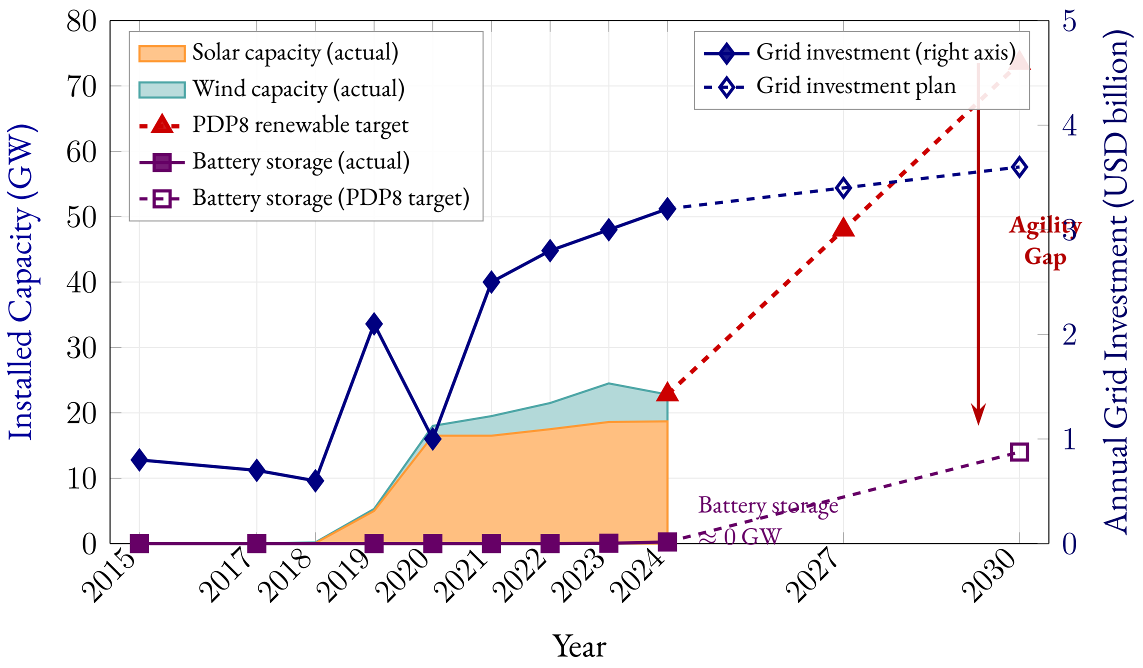

Figure 1.1: Descriptive evidence for the "agility gap." Left axis: installed renewable capacity (GW), decomposed into solar (orange) and wind (teal). Capacity surged from near-zero in 2017 to 22.8 GW by 2024; PDP8 targets 73.5 GW by 2030. Right axis: annual grid transmission investment (USD billion), which grew modestly from \$0.6–0.8B to \$3.2B. Battery storage (violet) remained near zero through 2024 despite a PDP8 target of 14 GW by 2030. Data: PDP8 (Decision 500/QD-TTg), EVN annual reports, IEA Vietnam Energy Outlook 2023, IRENA.

2 Literature Review

There has been a significant amount of environmental policy research utilizing DSGE models to analyze macroeconomic variables.[3][4] A central debate in E-DSGE research concerns the optimal instrument for emission control under uncertainty. Fischer and Springborn (2011) and Bosetti and Maffezzoli (2013) argue that car-

[3] See Annicchiarico and Di Dio (2015), Dissou and Karnizova (2016), Annicchiarico and Di Dio (2017), Khan et al. (2019), Chan (2020a), Chan (2020b), Lintunen and Vilmi (2021), and Xiao et al. (2022).

[4] Another body of literature that focuses on estimating the Social Cost of Carbon (SCC), called Integrated Assessment Models (IAM). Key works include Nordhaus's DICE/RICE (Nordhaus, 1992, 1994), Hope's PAGE (Hope, 2006), Tol's FUND (Tol, 2002), and advanced IAMs like WITCH (Bosetti et al., 2006), REMIND (Bauer et al., 2013), and GCAM (Calvin et al., 2019). IAMs face criticism regarding discounting and damage function assumptions (Weitzman, 2009; Pindyck, 2017). For a review, see Fernández-Villaverde et al. (2025).

bon taxes generally outperform quantity-based instruments (like cap-and-trade) by reducing volatility in critical macroeconomic indicators and effectively lowering emissions. However, other scholars emphasize that these policies cannot be static; Heutel (2012) demonstrates that optimal environmental policy must adjust pro-cyclically with the business cycle to mitigate recessionary impacts. More recently, attention has shifted to how these policies interact with structural frictions. For instance, Chan and Punzi (2023) reveals that the complementarity between energy and capital amplifies the transmission of shocks, suggesting that welfare-maximizing carbon taxes must be calibrated lower when endogenous capital utilization is taken into account.

Recently, Airaudo et al. (2025) advanced the Environmental DSGE literature by introducing a New Keynesian small open economy model that captures the macroeconomic costs of the green transition. Their analysis reveals that carbon taxes reduce brown energy use but cause short-term inflation and output losses, a phenomenon they term "greenflation", due to limited short-run substitutability between energy and traditional inputs. Crucially, they explicitly model public green investment as a fiscal instrument that can boost the productivity of the green sector. Their findings suggest that while carbon taxes alone are contractionary, mixed strategies that recycle carbon revenues into green public investment yield comparable environmental outcomes at lower welfare costs.

While Airaudo et al. (2025) focus on the short-run frictional costs of carbon taxation and fiscal policy coordination, this capstone complements their analysis by examining a distinct medium-run friction: the reliability penalty arising from renewable intermittency when flexibility infrastructure lags deployment. Whereas their "greenflation" emerges from incomplete factor substitutability under carbon pricing, the present model identifies an additional channel—endogenous grid instability—that operates even absent explicit carbon taxes, driven instead by the physical characteristics of VRE integration. This insight directly informs the policy discussion in Section 8.

While Airaudo et al. (2025) highlight the short-run frictional costs of transition, a distinct strand of literature rooted in endogenous growth theory offers a mechanism for long-run technological progress through innovation. The idea that factor scarcity directs innovation toward the scarce factor dates to Hicks (1932), who observed that a rise in the relative cost of a factor creates an inducement to discover methods of economizing on it. This hypothesis was formalized in the induced innovation literature by Kennedy (1964) and Drandakis and Phelps (1966), who showed within neoclassical growth models how equilibrium factor prices systematically bias the direction of technical progress. The modern Directed Technical Change (DTC) framework, pioneered by Acemoglu (2002) and applied to energy and environment by Acemoglu et al. (2012), models innovation as a resource allocation problem where scientists choose which sector to develop. This framework highlights two competing forces: the market size effect, which encourages innovation in the larger sector, and the price effect, which directs innovation toward the sector with higher prices. In the standard model, the market size effect dominates, locking the economy into dirty technologies. However, Acemoglu et al. (2012) show that policy intervention can manipulate these effects to redirect innovation toward clean technologies. Closest to the present paper, Hassler et al. (2021) develop a DTC model in which natural resource scarcity endogenously redirects technological change away from energy-intensive inputs — demonstrating that physical scarcity, not just price signals, can serve as the directing force for innovation.

This theoretical mechanism has found robust empirical support. Popp (2002) provided early evidence that higher energy prices significantly induce energy-efficient patenting activity. More recently, Aghion et al. (2016) utilized microdata from the global automotive industry to show that firms tend to innovate in areas where they have prior expertise (path dependence), but that fuel price shocks successfully redirect R&D toward electric and hybrid technologies. Similarly, Calel and Dechezleprêtre (2016) finds that firms regulated under the EU Emissions Trading System (ETS) increased low-carbon patenting by 10% compared to a matched control group without crowding out other R&D, suggesting that policy can effectively steer the direction of technical progress.

However, a fundamental limitation in standard DTC models is the assumption regarding the elasticity of substitution. The AABH framework relies on the premise that clean and dirty inputs are high substitutes. Under this condition, innovation in clean energy allows it to seamlessly replace fossil fuels. Yet, structural analyses challenge this assumption in the presence of intermittency. As renewable penetration rises, the lack of dispatchability renders clean energy an imperfect substitute for baseload fossil generation (Hirth, 2013; Joskow, 2011). Consequently, models that treat clean capital as a homogeneous block overlook the necessity of complementary innovations. As argued by Hart (2019), when inputs are complements rather than substitutes, technological progress must target not just generation, but the bottleneck technologies—specifically storage—that enable substitutability.

An alternative mechanism for technological progress is learning-by-doing, where costs decline through accumulated production experience rather than explicit R&D allocation. Arrow (1962) first formalized this idea, and it has been widely applied to renewable energy technologies. Nemet (2006) documents dramatic cost reductions in solar photovoltaics driven by cumulative deployment, while Kittner et al. (2017) shows similar patterns for battery storage using a two-factor learning curve that integrates both deployment experience and innovation activity. Way et al. (2022) extend this approach with stochastic Wright's Law forecasts across fifty technologies, projecting continued steep cost declines for batteries through 2050. Unlike DTC, learning-by-doing does not require modeling innovation sectors or resource constraints, making it particularly suitable for capturing technological progress in capital goods like batteries where costs decline through manufacturing scale and operational experience.

To address these dynamics, recent contributions have attempted to internalize these physical constraints within general equilibrium frameworks. For instance, Schreiner and Madlener (2020) develop a DSGE model for Germany that treats power grid infrastructure as a flexibility option required to bridge the mismatch between variable renewable supply and demand. Crucially, their simulations reveal that while grid investments increase reliability, the associated costs can lead to long-term decreases in GDP and employment unless accompanied by significant efficiency gains. This underscores that solving the intermittency problem is not merely a matter of capital accumulation (building more lines) but requires targeted technological progress in intertemporal substitution (storage).

While the engineering literature has long established the necessity of storage for renewable integration, macroeconomic frameworks have lagged in formally modeling this investment channel. Techno-economic analyses, such as Zakeri and Syri (2015) and Lund et al. (2015), demonstrate that, as renewable penetration increases, the primary cost driver shifts from generation (LCOE) to flexibility (integration costs). They argue that, without storage, the social cost of intermittency rises non-linearly due to the need for redundant fossil-fuel backup.

Despite this, the standard E-DSGE literature largely omits storage as an endogenous decision variable. Major recent contributions, including Batten and Millard (2020) and Annicchiarico et al. (2021), focus predominantly on pricing instruments (carbon taxes) or quantity restrictions (caps), effectively assuming that energy markets clear instantaneously. When backup is modeled, as in Schreiner and Madlener (2020), it is often treated as a static infrastructure cost or fossil generation, which inherently reintroduces carbon emissions or capital rigidities that dampen the efficacy of green transition policies.

This creates a significant modeling gap. By treating backup as fossil generation rather than intertemporal substitution, existing models fail to capture the capital-intensive nature of grid stability. As noted by Kittner et al. (2017), the declining costs of battery technologies imply that energy storage is transitioning from a niche ancillary service to a core infrastructure asset. Consequently, accurately capturing the "greenflation" trade-off requires a framework where agents can optimize investment not just in productive capital, but also in smoothing capital (BESS), a structural feature this capstone seeks to integrate.

Beyond the storage modeling gap, international trade emerges as a critical transmission channel for small open economies navigating the green transition. Open-economy macroeconomic research shows that energy price shocks behave like terms-of-trade shocks and can generate persistent effects on output, inflation, and external balances when a country is heavily dependent on imported energy (Obstfeld and Rogoff, 1995; Corsetti et al., 2010). In the environmental domain, a growing body of open-economy E-DSGE and two-country DSGE studies demonstrates that integrating carbon pricing across borders induces spillovers in trade flows, alters the competitiveness of carbon-intensive sectors, and raises carbon leakage risks (Böhringer et al., 2018; Balke and Winkler, 2022). These mechanisms matter quantitatively: recent guidance by the Network for Greening the Financial System (NGFS, 2022) and the IMF (International Monetary Fund, 2021) emphasizes that external exposure, particularly dependence on imported energy and intermediate goods, amplifies transition-related price shocks and materially affects domestic inflation and output under decarbonization scenarios.

For Vietnam specifically, trade linkages are not peripheral but central to its green-transition challenge. The country's rapid export-oriented industrialization has been accompanied by rising dependence on imported thermal coal and liquefied natural gas (LNG), making global fossil-fuel price fluctuations a direct source of domestic production-cost volatility (International Energy Agency, 2023; General Statistics Office of Vietnam, 2024). At the same time, Vietnam's energy-intensive export sectors, such as steel, cement, and fertilizers, face increasing exposure to foreign climate policies like the EU Carbon Border Adjustment Mechanism (European Commission, 2023). Empirical work on Vietnam and comparable emerging economies shows that trade openness interacts strongly with renewable-energy adoption and CO_2 emissions, reinforcing the need to jointly consider trade and environmental policy in macro-modeling (Al-Mulali et al., 2015; Vo et al., 2020). The broader literature on carbon leakage, border adjustments, and greenflation also underscores the relevance of modelling international connections: unilateral carbon pricing, carbon-border tariffs, or linked carbon markets produce measurable welfare, competitiveness, and terms-of-trade effects that an open-economy E-DSGE is well-suited to analyze (Fischer and Fox, 2012; Böhringer et al., 2022). These insights collectively highlight that, for a highly open economy like Vietnam, international trade in both energy and goods is not an auxiliary consideration but a defining structural feature of the green transition.

Taken together, the existing literature provides a strong foundation for environmental DSGE frameworks, yet it leaves critical blind spots regarding the interactions among intermittency, flexibility investment, and induced innovation. Building on these gaps, this capstone makes three core contributions. First, it introduces an endogenous reliability function that links capital utilization to the ratio of flexibility assets to renewable volatility, allowing intermittency to propagate through both the production function and the investment channel. Second, it incorporates a novel scarcity-induced technological change mechanism for Battery Energy Storage Systems (BESS), where technological progress responds endogenously to the reliability gap between target and actual grid performance. This mechanism combines the accumulation structure of Arrow's (1962) learning-by-doing tradition with the induced innovation logic of Hicks (1932), Kennedy (1964), Drandakis and Phelps (1966), and Acemoglu (2002): the reliability gap serves as the scarcity signal that directs innovative effort toward storage, while technology compounds from an inherited knowledge base in the manner of empirical battery experience curves (Kittner et al., 2017; Way et al., 2022). Crucially, the transmission of this scarcity signal is governed by a regulatory parameter χ, allowing the model to quantify how market distortions — such as Vietnam's administratively set tariffs and underdeveloped ancillary service markets — suppress technological progress even when grid stress is severe. Third, it distinguishes between agile private battery investment and slow-moving public grid infrastructure, capturing the structural *agility gap* observed in developing economies. By integrating these elements, the capstone argues that market-based scarcity signals are essential complements to physical storage investment in transforming the green transition from a cost burden into a sustainable growth path.

3 Methodology

3.1 Model Overview

I develop a Small Open Economy Dynamic Stochastic General Equilibrium (DSGE) model calibrated to Vietnam. The model is designed to analyze the macroeconomic consequences of renewable-energy intermittency when the accumulation of grid flexibility lags behind the expansion of variable renewable generation, while capturing the external balance implications through international borrowing and lending. The framework extends standard Environmental DSGE (E-DSGE) models by introducing two departures from the canonical structure.

First, grid reliability is endogenized as a technological constraint on effective capital services rather than treated implicitly through energy prices or exogenous productivity shocks. This reflects the physical reality that electricity systems with high renewable penetration are constrained by temporal mismatches between supply and demand. Second, I embed a scarcity-induced technological change mechanism that governs the rate of improvement in battery storage technologies. This allows the economy to respond endogenously to reliability scarcity through technological progress rather than relying solely on capital accumulation or policy intervention.

The economy consists of a representative household and a representative firm operating in a small open economy with access to international bond markets. Households supply labor, accumulate productive capital, and trade bonds with the rest of the world at a debt-elastic interest rate. Firms produce final output using a CES aggregator of value-added (from capital and labor) and a fixed energy endowment. Grid reliability is determined endogenously by the stock of flexibility assets (batteries and grid infrastructure) relative to renewable volatility. Battery technology improves through scarcity-induced innovation driven by reliability gaps, while public grid investment follows a rule-based response to reliability shortfalls.

A central modeling choice is the treatment of capital utilization. In contrast to standard RBC or New Keynesian models, capital utilization in this framework is not a choice variable and does not affect depreciation. Instead, it reflects the fraction of installed productive capital that is operational given the state of grid reliability. This interpretation aligns utilization with system-wide power availability rather than firm-level intensity of use.

3.2 Households

A representative household maximizes expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_t) - \varphi \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right] \tag{1}$$

subject to the budget constraint:

$$C_t + I_{p,t} + P_{bat,t} I_{bat,t} + B_t^* + \tau Y_t = w_t L_t + R_t K_{p,t-1} + (1 + r_{t-1}^*) B_{t-1}^* \tag{2}$$

where $I_{p,t}$ is productive capital investment, $I_{bat,t}$ battery investment at world price $P_{bat,t}$, B_t^* denotes net foreign asset holdings earning the country-specific interest rate r_t^*, τ is a fixed proportional tax on output levied by the government to finance public grid investment (specified in Section 3.6), w_t is the competitive real wage, and R_t is the rental rate of productive capital. The household rents out the capital stock carried into the period, $K_{p,t-1}$, so that capital income is $R_t K_{p,t-1}$. From the firm's profit maximization, $w_t = (1-\alpha)V_t/L_t$ and $R_t = \alpha V_t / K_{p,t-1}$, so that total factor income equals $w_t L_t + R_t K_{p,t-1} = (1-\alpha)V_t + \alpha V_t = V_t$.[5]

The spending item $I_{p,t}$ in the budget constraint represents the household's purchase of new productive capital goods. It is linked to the next-period capital stock through the standard perpetual-inventory accumulation law (equation (26)):

$$K_{p,t+1} = (1 - \delta_p) K_{p,t} + I_{p,t}$$

so that $I_{p,t} = K_{p,t+1} - (1-\delta_p) K_{p,t}$: investment is simply the increment to next period's capital stock above what survives depreciation from the current stock. Equivalently, the household's true intertemporal choice variable is the end-of-period capital stock $K_{p,t+1}$, not $I_{p,t}$ itself. In the Lagrangian derivation (Appendix D), $I_{p,t}$ is substituted out using this relation, so the optimisation is carried out over $\{C_t, L_t, K_{p,t+1}, B_t^*\}$. The Euler equation (3) below follows directly from the first-order condition with respect to $K_{p,t+1}$.

The household chooses $\{C_t, L_t, K_{p,t+1}, B_t^*\}$ to maximize utility subject to the budget constraint (2) and the accumulation law above. The first-order conditions are derived as follows.

[5]The timing convention here is standard in DSGE models with capital: capital installed by the end of period $t-1$ (i.e., $K_{p,t}$ in the accumulation equation $K_{p,t} = (1-\delta_p)K_{p,t-1} + I_{p,t}$) becomes productive in period t, earning rental rate R_t. Investment $I_{p,t}$ is a use of funds in period t but does not contribute to production until period $t+1$. Writing $R_t K_{p,t}$ on the income side of the budget constraint would double-count $I_{p,t}$, since $K_{p,t}$ already embeds it. Consequently, the value-added function is $V_t = Z(u_t K_{p,t-1})^\alpha L_t^{1-\alpha}$, using the predetermined capital stock $K_{p,t-1}$.

Euler equation for capital. With proportional taxation, the after-tax return to capital in period $t+1$ is $(1-\tau)\alpha Y_{t+1}/K_{p,t}$, where τ is the fixed tax rate. The intertemporal optimality condition becomes:[6]

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{(1-\tau)\,\alpha Y_{t+1}/K_{p,t} + 1 - \delta_p}{C_{t+1}} \right] \tag{3}$$

The household invests in capital until the marginal cost of saving today equals the expected discounted after-tax return. Since τ is constant, the only dynamic distortion comes from the standard Euler mechanism: a fall in expected future output Y_{t+1} (due to persistent intermittency shocks) lowers the after-tax return to capital, compressing investment. The constant tax wedge shifts the steady-state capital stock downward but does not amplify the shock response beyond this RBC channel.

Labor supply. With the proportional tax, the household equates the marginal disutility of labor to the after-tax marginal utility of income. Substituting the competitive wage $w_t = (1-\alpha)V_t/L_t$:[7]

$$(1-\tau)(1-\alpha)V_t = \varphi\, C_t\, L_t^{1+\sigma_L} \tag{4}$$

The after-tax labor share of value-added (left side) equals the household's required compensation for working. The proportional tax τ drives a wedge that reduces effective labor income, lowering equilibrium labor supply relative to the lump-sum benchmark. The parameter φ is calibrated endogenously to match steady-state labor hours of $L_{ss} = 0.33$.

3.3 The Production Sector

3.3.1 Final Goods Production

Final output Y_t is produced by a representative firm using a nested Constant Elasticity of Substitution (CES) technology that combines value-added and energy inputs. To reduce notational clutter in the CES expressions that follow, I adopt the convention of writing exponents in terms of the *substitution parameter* $s \equiv (\sigma - 1)/\sigma$

[6] The full step-by-step Lagrangian derivation of this condition is provided in Appendix D.
[7] The full step-by-step Lagrangian derivation of this condition is provided in Appendix E.

for a given elasticity σ. When $\sigma < 1$ (complements), $s < 0$; when $\sigma > 1$ (substitutes), $s > 0$; when $\sigma = 1$ (Cobb-Douglas), $s = 0$. Using $s_E \equiv (\sigma_E - 1)/\sigma_E$ for the output aggregator:

$$Y_t = \left[(1-\omega_E)V_t^{s_E} + \omega_E \bar{E}^{s_E}\right]^{1/s_E} \tag{5}$$

Output is a weighted aggregate of value-added V_t and energy E_t, where neither input can fully substitute for the other when $\sigma_E < 1$: shutting down either drives output toward zero, making energy and productive activity essential complements.

This formulation follows the environmental macroeconomics literature (Heutel, 2012; Airaudo et al., 2025) and captures the limited short-run substitutability between energy and non-energy inputs. When the elasticity of substitution σ_E is low (calibrated at 0.6), disruptions to energy supply translate into disproportionately large output losses, a feature that is empirically relevant for energy-intensive emerging economies.

Value-added production combines labor and effective capital services:

$$V_t = Z\left(u_t K_{p,t-1}\right)^{\alpha} L_t^{1-\alpha} \tag{6}$$

Value-added is produced from "effective capital" $u_t K_{p,t-1}$—the fraction of *previously installed* capacity that is actually operational—combined with labor. A factory with machines that cannot run due to power outages produces less, even if the machines physically exist. The timing $K_{p,t-1}$ reflects the standard capital-in-advance convention: capital installed by the end of period $t-1$ is the stock available for production in period t (see footnote 5).

Here, $K_{p,t-1}$ is the physical stock of productive capital carried into period t, and $u_t \in (0,1)$ represents the fraction of that stock that can be operated given grid conditions. Unlike standard utilization models (e.g., Greenwood et al. (1988)), u_t is not chosen optimally by firms and does not influence capital depreciation. Instead, it captures system-wide outages, curtailment, and power shortages that prevent installed capital from being used. In this sense, grid instability manifests as an endogenous, state-dependent productivity wedge: a 3% reliability shortfall ($u_t = 0.97$) is equivalent to a TFP loss of approximately $\alpha \times 3\% \approx 1\%$.

In the implemented model, energy enters the CES aggregator as a fixed endowment $\bar{E}$, reflecting the assumption that total energy supply is determined by installed capacity and policy targets (PDP8) rather than optimized period-by-period.

This simplification allows the model to focus on the key channel of interest—how renewable intermittency affects grid reliability and, through it, effective capital utilization—without requiring a full energy market clearing block. The intermittency shock enters through the reliability function (equation (7) below) rather than through energy supply directly.

The household's optimality conditions—the capital Euler equation (3) and labor supply condition (4), as derived in Section 3.2—together with the production technology determine the equilibrium allocation of capital and labor.

3.4 Grid Reliability and Flexibility Assets

3.4.1 Reliability Constraint

Grid reliability is modeled as a reduced-form approximation to the probability that aggregate electricity demand can be met without load shedding. The functional form is motivated by the standard reliability engineering framework, where the Loss-of-Load Probability (LOLP) declines exponentially as reserve margins (here, flexibility assets) increase relative to demand uncertainty (here, renewable volatility). In engineering practice, LOLP is defined as the probability that available generation falls short of demand in a given period; the complement $1 - \text{LOLP}$ measures the fraction of time the system operates reliably. The exponential form $u = 1 - \exp(-z)$ is the simplest functional form that maps a non-negative "adequacy ratio" $z \geqslant 0$ into a probability bounded on $(0, 1)$, while preserving the key engineering property that additional reserves exhibit diminishing marginal returns to reliability. This is the same mathematical structure underlying the Poisson reliability model widely used in power systems analysis (Zakeri and Syri, 2015; Lund et al., 2015).

The reliability constraint is formalized as:

$$u_t = 1 - \exp\left(-\psi \frac{F_t}{\phi_{int} \cdot \text{Vol}_{ren,t}}\right) \tag{7}$$

Grid reliability is an increasing, concave function of the ratio of flexibility assets (F_t) to renewable volatility ($\text{Vol}_{ren,t}$). When the system has abundant flexibility relative to volatility, nearly all capital can operate ($u_t \approx 1$); when flexibility is scarce, outages proliferate and utilization collapses.

The variables and parameters are:

- F_t is the stock of flexibility assets (batteries + grid infrastructure, defined below)
- $\phi_{int} > 0$ is a scaling parameter that converts renewable volatility into effective flexibility requirements; it captures the complexity of integrating variable renewables into the grid (calibrated endogenously to ≈ 55.6)
- $\text{Vol}_{ren,t}$ measures renewable output volatility (strictly positive)
- $\psi > 0$ governs the sensitivity of reliability to the flexibility-to-volatility ratio (calibrated at 2.0)

This exponential formulation ensures that:

1. When $F_t \to \infty$, utilization approaches unity (perfect reliability): $u_t \to 1$
2. When $F_t \to 0$, utilization approaches zero (grid collapse): $u_t \to 0$
3. The function is always bounded: $u_t \in (0, 1)$ for all $F_t > 0$
4. Marginal returns to flexibility are strictly diminishing
5. The reliability penalty increases exponentially with the volatility-to-flexibility ratio

This exponential functional form is standard in reliability engineering (Zakeri and Syri, 2015; Lund et al., 2015) and ensures that flexibility requirements rise nonlinearly as VRE penetration increases, capturing the convex integration costs documented in the empirical literature.

To illustrate the nonlinearity, Table 3.1 reports the reliability response to flexibility shortfalls under the baseline calibration. The convex curvature is evident: a 10% flexibility deficit reduces reliability by 1.26 percentage points (pp), but a 50% deficit causes a 14.32 pp collapse—more than 11 times the proportional response. This accelerating penalty captures the engineering reality that the first units of flexibility (spinning reserves, fast-ramping gas) are cheap and effective, while maintaining reliability under extreme shortfalls requires exponentially more resources.

Table 3.1: **Reliability Response to Flexibility Shortfalls**

Flexibility Shortfall	Reliability (u_t)	Rel. Loss (pp)	Approx. Output Loss
0% (steady state)	97.00%	—	—
-5%	96.43%	0.57	-0.21%
-10%	95.74%	1.26	-0.45%
-20%	93.95%	3.05	-1.10%
-30%	91.41%	5.59	-2.02%
-50%	82.68%	14.32	-5.17%

Note: Computed from equation (7) under the baseline calibration ($\psi = 2.0, \phi_{int} = 55.6$). Output loss approximated as $\alpha \times \Delta u/u_{ss}$, capturing the direct effective-capital channel. The convexity of the reliability function implies that moderate flexibility shortfalls (10–20%) have manageable consequences, but large deficits trigger disproportionate reliability collapses—motivating the "Reliability Valley" analysis in Section 6.

The volatility term captures the absolute variability of renewable generation:

$$\overline{\text{Vol}}_{ren} = \theta_{ren} \cdot K_{ren} \cdot \sigma_{ren} \tag{8}$$

where θ_{ren} is the renewable penetration share (dimensionless), K_{ren} is installed renewable capacity (GW), and σ_{ren} is the coefficient of variation of the intermittency shock (dimensionless). As defined, $\overline{\text{Vol}}_{ren}$ carries units of GW. Dimensional consistency in the reliability function (7) is preserved by the scaling parameter ϕ_{int}, which has units of GW per unit of flexibility stock, so that the ratio $F_t/(\phi_{int} \cdot \text{Vol}_{ren,t})$ is dimensionless as required by the exponential.

In the quantitative implementation, $\overline{\text{Vol}}_{ren}$ is treated as a **calibrated parameter** rather than an endogenous variable, set to match the baseline renewable fleet characteristics under PDP8. This model deliberately treats the trajectory of Variable Renewable Energy (VRE) deployment as exogenous: firms do not dynamically optimize solar or wind capacity; instead, VRE follows the government's PDP8 plan implicitly. This simplification isolates the endogenous response of flexibility assets—the "agility gap"—and is a natural starting point before fully endogenizing VRE investment.

The intermittency shock $\varepsilon_{ren,t}$ enters the reliability function (7) multiplicatively as $\overline{\text{Vol}}_{ren} \cdot \exp(\varepsilon_{ren,t})$, generating stochastic variation in effective volatility around the calibrated baseline.

Conceptually, $\overline{\text{Vol}}_{ren}$ varies with the size of the renewable fleet, capturing the empirical finding that integration costs rise with penetration (Hirth, 2013). In the deterministic transition simulations (Section 6.4), $\overline{\text{Vol}}_{ren}$ is allowed to increase along the policy-determined VRE expansion path.

3.4.2 Public and Private Flexibility Assets

Total system flexibility F_t is produced by combining two distinct assets: private battery storage and public grid infrastructure. These assets serve complementary roles. Battery storage shifts energy *across time* (charging when supply exceeds demand, discharging when it falls short), while grid infrastructure shifts energy *across space* (transmitting power from surplus to deficit regions). Using the shorthand $s_F \equiv (\rho - 1)/\rho$ for the flexibility substitution parameter (where ρ is the elasticity of substitution between battery and grid capital), the flexibility bundle is:

(9)
$$F_t = \left[\mu(A_{bat,t-1}K_{b,t-1})^{s_F} + (1-\mu)K_{g,t-1}^{s_F}\right]^{1/s_F}, \qquad \rho < 1 \;\Leftrightarrow\; s_F < 0$$

Flexibility is a weighted combination of technology-augmented battery capacity ($A_{bat,t-1}K_{b,t-1}$) and grid infrastructure ($K_{g,t-1}$), where neither can fully replace the other. A grid with abundant batteries but no transmission lines cannot deliver stored power; a grid with extensive transmission but no storage cannot smooth intermittent supply. The negative exponent $s_F < 0$ (since $\rho = 0.4 < 1$) ensures that flexibility collapses if either input approaches zero—a mathematical expression of this engineering constraint.

Timing convention. All three inputs to the flexibility aggregator are predetermined: capital stocks $K_{b,t-1}$ and $K_{g,t-1}$ installed by the end of period $t-1$, and battery technology $A_{bat,t-1}$ induced in period $t-1$. Scarcity-induced innovation in period t updates $A_{bat,t}$ (equation 16), but this improvement only enters flexibility in period $t+1$. This one-period-delay convention means that innovative effort is directed today but becomes operationally productive tomorrow, eliminating any within-period simultaneity between the innovation and reliability equations.

The elasticity of substitution $\rho = 0.4$ is calibrated below unity to enforce this complementarity. The parameter $\mu = 0.16$ governs the relative weight of battery storage in the aggregate, reflecting Vietnam's current early-stage storage deployment. Innovation affects flexibility through $A_{bat,t-1}$: technology improvements induced by last period's reliability gap raise the usable energy-shifting capacity per unit of installed storage in the current period.

The complementarity parameter $\rho < 1$ has a precise formal consequence for the returns to coordinated flexibility investment. See Proposition 2 in Appendix B for the statement and quantitative characterization.

3.4.3 Laws of Motion

Public grid capital evolves according to a standard accumulation equation:

$$K_{g,t+1} = (1 - \delta_g)K_{g,t} + I_{grid,t} \tag{10}$$

Grid investment is undertaken by the government and is subject to institutional inertia, reflecting planning delays and construction lags. As a result, public flexibility adjusts slowly to changes in renewable penetration.

Private battery capital evolves as:

$$K_{b,t+1} = (1 - \delta_b)K_{b,t} + I_{bat,t} \tag{11}$$

Battery capital follows a standard linear accumulation law without adjustment costs, as the modular and scalable nature of battery storage allows rapid deployment without the installation frictions typically associated with heavy infrastructure.

Battery Investment Decision. A fully endogenous, forward-looking optimization of battery capital—subject to the highly non-linear exponential reliability constraint—would create non-convexities that complicate the Blanchard-Kahn conditions required for local approximation. The approach here instead derives a reduced-form investment rule that proxies the gradient of the structural shadow price while keeping the system computationally tractable and ensuring local saddle-path stability.

To see where this rule comes from, consider the shadow rental rate on storage services. The marginal value of an additional unit of battery capital flows through the chain $K_b \to F \to u \to V \to Y$:

$$R_{b,t} \equiv \frac{\partial Y_t}{\partial F_t} \cdot \frac{\partial F_t}{\partial (A_{bat,t-1}K_{b,t-1})} \cdot A_{bat,t-1} \tag{12}$$

Each component is computed from the model's structural equations. From the CES flexibility aggregator (9):

$$\frac{\partial F_t}{\partial(A_{bat,t-1}K_{b,t-1})} = \mu\left(\frac{A_{bat,t-1}K_{b,t-1}}{F_t}\right)^{-1/\rho_f} \tag{13}$$

The marginal product of flexibility in output operates through the reliability channel. Combining the exponential reliability function (7) with the value-added function (6) and the CES output aggregator (5):

$$\frac{\partial Y_t}{\partial F_t} = \underbrace{\frac{\partial Y_t}{\partial V_t}}_{\text{CES share}} \cdot \underbrace{\frac{\partial V_t}{\partial u_t}}_{\text{capital utilization}} \cdot \underbrace{\frac{\partial u_t}{\partial F_t}}_{\text{reliability gradient}} = \frac{\partial Y_t}{\partial V_t} \cdot \alpha\frac{V_t}{u_t} \cdot \frac{\psi(1-u_t)}{\phi_{int} \cdot \mathrm{Vol}_{ren,t}} \tag{14}$$

where $\partial u_t/\partial F_t = \psi(1-u_t)/(\phi_{int} \cdot \mathrm{Vol}_{ren,t})$ follows from differentiating the exponential reliability function, and $\partial V_t/\partial u_t = \alpha V_t/u_t$ from the Cobb-Douglas value-added specification. Note that $R_{b,t}$ is increasing in the reliability gap $(1-u_t)$: when reliability falls below target, the marginal return to storage rises sharply due to the exponential curvature of the reliability function.

Table 3.2 evaluates each component of the chain at steady-state values, yielding a shadow rental rate of $R_{b,ss} = 10.93\%$ quarterly. The decomposition reveals that the *flexibility bottleneck*—the large marginal product of batteries in the CES aggregator ($\partial F/\partial(A_{bat}K_b) = 2.04$)—is the dominant force, reflecting the scarcity of storage relative to grid capital. The reliability gradient ($\partial u/\partial F = 0.20$) is moderate because the system operates near the flat portion of the exponential curve at $u_{ss} = 0.97$.

Table 3.2: **Marginal Product Decomposition of Battery Rental Rate ($R_{b,ss}$)**

Component	**Expression**	**Value**	**Interpretation**
CES output share	$\partial Y/\partial V$	0.725	Value-added dominates output
Utilization effect	$\partial V/\partial u = \alpha V/u$	0.370	Capital utilization channel
Reliability gradient	$\partial u/\partial F$	0.200	Moderate (near flat of exp. curve)
Flexibility bottleneck	$\partial F/\partial(A_{bat}K_b)$	2.040	Scarce factor in CES
Shadow rental rate	$R_{b,ss}$	**0.109**	**10.93% quarterly**

Note: Evaluated at the deterministic steady state. $R_{b,ss} = (\partial Y/\partial V) \times (\partial V/\partial u) \times (\partial u/\partial F) \times (\partial F/\partial(A_{bat}K_b)) \times A_{bat} = 0.725 \times 0.370 \times 0.200 \times 2.040 \times 1.0 = 0.109$. The large flexibility bottleneck term reflects batteries' position as the scarce factor ($\mu = 0.16$) in the CES aggregator. When u_t falls below target, the reliability gradient steepens (since $\partial u/\partial F \propto (1 - u_t)$), amplifying R_b and triggering the investment response.

The rental rate $R_{b,t}$ depends on the reliability gradient $\partial u/\partial F = \psi(1 - u_t)/(\phi_{int} \cdot \text{Vol}_{ren,t})$, which is proportional to the reliability gap $(1 - u_t)$. When reliability falls below target, $(1 - u_t)$ rises, steepening the gradient and raising $R_{b,t}$ above its steady-state value. The ratio u^*/u_t is a monotone transformation of this gap: at steady state $u_t = u^* = 0.97$ and the ratio equals unity; when u_t drops below u^*, the ratio rises, driving investment above replacement levels. This motivates the reduced-form investment rule:

$$\frac{I_{bat,t}}{\bar{I}_{bat}} = \left(\frac{u^*}{u_t}\right)^{\phi_{grid}} \cdot \frac{1}{P_{bat,t}} \tag{15}$$

The exponent ϕ_{grid} governs the elasticity of the investment response to the reliability gap—higher values imply more aggressive battery deployment when the grid is stressed. The price term $1/P_{bat,t}$ reflects the cost side: Vietnam is a price-taker in global battery markets, so higher world battery prices reduce real investment even when the return to storage is high. At steady state, the rule collapses to $I_{bat,ss}/\bar{I}_{bat} = 1$, consistent with capital replacement at $I_{bat,ss} = \delta_b K_{b,ss}$.

3.5 Scarcity-Induced Technological Change in Battery Storage

Technological progress in battery storage is modeled as an endogenous process driven by the reliability gap — the gap between the grid's target and actual performance. This mechanism combines two distinct strands of the innovation literature. The *accumulation structure* follows Arrow's (1962) learning-by-doing tradition: technology compounds multiplicatively from an inherited knowledge base $A_{bat,t-1}$, with bounded and mean-reverting growth consistent with empirical learning curves (Nemet, 2006; Kittner et al., 2017). The *direction* of innovation, however, follows the induced innovation framework of Hicks (1932), Kennedy (1964), and Drandakis and Phelps (1966), as formalized in the Directed Technical Change (DTC) literature by Acemoglu (2002): when a factor is scarce, the private return to improving it rises, directing technological effort toward that bottleneck. Here the relevant scarcity signal is grid reliability itself — when the grid is stressed, the economic value of better battery technology spikes, inducing accelerated innovation in storage. The evolution of battery technology follows:

$$A_{bat,t} = \underbrace{A_{bat,t-1}}_{\text{inherited knowledge}} \times \left(1 + \underbrace{\eta_{bat}}_{\text{innovation elasticity}} \cdot \underbrace{\chi}_{\substack{\text{signal} \\ \text{transmission}}} \cdot \underbrace{\frac{u^* - u_t}{u^*}}_{\text{reliability gap}}\right) \tag{16}$$

where:

- $\eta_{bat} > 0$ is the technology adoption elasticity governing how strongly the reliability scarcity signal translates into innovation
- $\chi \in (0, 1]$ captures regulatory distortions in price signals
- u^* is the target reliability level
- u_t is the current reliability

This specification ensures that technological progress is zero when reliability is at target ($u_t = u^*$) and accelerates proportionally when reliability falls below target. The mechanism is distinct from both pure learning-by-doing (where the driver is cumulative deployment or production volume) and pure Directed Technical Change (where scientists allocate R&D effort across sectors). Instead, it captures the empirically documented phenomenon that factor scarcity raises the private return to innovation in that factor, directing technological capabilities toward the binding bottleneck (Popp, 2002; Acemoglu et al., 2012; Aghion et al., 2016; Hassler et al., 2021). The accumulation structure — compounding from $A_{bat,t-1}$ with bounded growth — ensures that improvements persist and exhibit the saturation properties documented in battery experience curves (Kittner et al., 2017; Way et al., 2022).

Timing and causal ordering. The update $A_{bat,t}$ is determined within period t from the current reliability gap, but it only enters the flexibility aggregator (9) in period $t+1$ as $A_{bat,t-1}$. This one-period delay means the sequence within any period t is strictly recursive and free of simultaneity: (i) F_t is computed from predetermined $A_{bat,t-1}$ and $K_{b,t-1}$; (ii) u_t follows from F_t and the current intermittency shock; (iii) $A_{bat,t}$ is then updated from the reliability gap $(u^* - u_t)/u^*$. Economically, this captures the reality that scarcity signals are processed and innovation decisions are made within the current quarter, but the resulting improvements to battery systems become operationally available only in the following period.

The parameter χ captures regulatory distortions that dampen the transmission of reliability scarcity into private incentives. In regulated markets with incomplete remuneration for ancillary services (e.g., fixed tariffs, lack of capacity payments, or missing real-time pricing), $\chi < 1$, weakening innovation responses even when flexibility is scarce. In fully liberalized markets with scarcity pricing, $\chi = 1$.

Policy Implication for Vietnam: In the specific context of Vietnam, the state utility (EVN) currently operates with strictly regulated retail tariffs and lacks a mature Competitive Wholesale or Retail Electricity Market (VWEM/VREM) tailored for dynamic capacity pricing. Therefore, Vietnam's current regulatory baseline corresponds to a χ significantly below 1 (close to 0). This formalizes a critical policy recommendation: to unlock the scarcity-induced innovation channel for battery storage, Vietnam must accelerate its transition toward competitive markets featuring time-of-use (ToU) and capacity payments. Without these scarcity signals, technological progress will stagnate regardless of the physical need for flexibility.

This formulation differs from Directed Technical Change in three key respects. First, there is no R&D sector or explicit allocation of scientists. Second, innovation is directed by the *reliability gap*—a physical grid signal—rather than by intersectoral price competition. Third, technological progress compounds from an inherited knowledge base, capturing the path-dependence and saturation properties of empirical battery learning curves (Kittner et al., 2017; Way et al., 2022). The χ parameter formalizes the empirical finding that policy-induced scarcity signals—such as carbon markets or capacity payments—are necessary to fully activate the innovation response (Calel and Dechezleprêtre, 2016; Aghion et al., 2016). Underlying all three investment and innovation channels is a single structural property of the exponential utilization function: its marginal effect of flexibility on reliability,

$$\frac{\partial u_t}{\partial F_t} = (1 - u_t) \cdot \frac{\psi}{\phi_{int} \cdot \text{Vol}_{ren,t}}, \tag{17}$$

is proportional to the current reliability shortfall $(1 - u_t)$. When reliability is already high, additional flexibility yields diminishing returns; when the grid is stressed, flexibility becomes critically valuable. The battery investment rule (15) and the innovation equation (16) both respond to this scarcity through the ratio u^*/u_t, which amplifies each response precisely when the shortfall is largest.

3.6 Government

The government does not solve a Ramsey optimization problem. Instead, it follows an institutional reliability-targeting rule motivated by Vietnam's Power Development Plan 8 (PDP8). Public grid investment responds mechanically to deviations of reliability from a target level:

$$\frac{I_{grid,t}}{\bar{I}_{grid}} = \left(\frac{u^*}{u_t}\right)^{\phi_{grid}} \exp(\varepsilon_{I,t}) \tag{18}$$

The government spends more on grid infrastructure when reliability falls below target, and less when the grid is performing well. This rule is structurally identical to the battery investment rule (15) except that it lacks a price term (grid investment uses domestically produced goods) and includes an implementation shock $\varepsilon_{I,t}$ reflecting bureaucratic delays, procurement bottlenecks, and construction lags.

This rule captures the reactive nature of infrastructure policy, where investment accelerates following reliability crises rather than anticipating them. The parameter ϕ_{grid} governs the aggressiveness of the fiscal response.

The government finances grid investment through a proportional tax τ on household output income. The tax rate is a *fixed parameter*, calibrated in steady state to satisfy the government's budget constraint:

$$\tau_{ss} = \frac{I_{grid,ss}}{Y_{ss}} = \frac{\delta_g K_{g,ss}}{Y_{ss}} = 0.01425 \tag{19}$$

so that in steady state, tax revenue τY_{ss} exactly covers replacement grid investment $I_{grid,ss}$. This proportional tax is economically motivated by Vietnam's practice of financing EVN grid expenditures through regulated electricity tariffs, which act as a charge roughly proportional to economic activity. Unlike a lump-sum tax, τ is distortionary: it drives a permanent wedge between the household's gross and net income, reducing the effective return to labor and capital and thereby shifting equilibrium allocations relative to the lump-sum benchmark. Since τ is fixed, it does not respond to cyclical fluctuations in grid investment — it acts as a constant fiscal wedge that affects the steady-state level of capital and output but does not amplify business cycle dynamics.

3.7 Battery Price Dynamics

Battery investment goods are priced at world price $P_{bat,t}$, which enters the resource constraint and the battery investment rule. The world price of batteries evolves according to a persistent AR(1) process:

$$\log P_{bat,t} = \rho_{bat} \log P_{bat,t-1} + \varepsilon_{bat,t} \tag{20}$$

where $\varepsilon_{bat,t} \sim N(0, \sigma^2_{bat})$ captures global shocks to battery costs (e.g., lithium price volatility, supply chain disruptions). This channel allows the model to examine how external cost shocks to the green transition propagate through the domestic economy. In steady state, $P_{bat,ss} = 1$ (normalized).

3.8 External Sector

The model is embedded in a small open economy framework following the debt-elastic interest rate premium approach of Schmitt-Grohé and Uribe (2003). The representative household has access to international bond markets, where it can borrow or lend at a country-specific interest rate that depends on the economy's net foreign asset position.

3.8.1 Bond Euler Equation

The household's optimality condition for international bond holdings yields:

$$\frac{1}{C_t} = \beta\,\mathbb{E}_t\left[\frac{1+r_t^*}{C_{t+1}}\right] \tag{21}$$

The household lends abroad until the cost of forgoing one unit of consumption today equals the expected discounted return from the foreign bond. This is analogous to the capital Euler equation (3), but with the foreign interest rate replacing the domestic marginal product of capital. Combining equations (3) and (21) yields the no-arbitrage condition linking domestic and external returns:

$$(1-\tau)\,\alpha Y_{t+1}/K_{p,t} + 1 - \delta_p = 1 + r_t^* \tag{22}$$

This is not an independent equation but rather follows directly from the two Euler conditions. It is listed here to clarify the economic intuition: in equilibrium, the domestic marginal product of capital equals the external borrowing rate.

3.8.2 Current Account and Budget Constraint

Combining the household budget constraint (2) with the government's financing identity (tax revenue τY_t funds grid investment $I_{grid,t}$, with the difference covered or built up in the foreign bond position) yields the economy-wide resource constraint. The proportional tax does *not* satisfy Ricardian equivalence: households optimise over after-tax income $(1-\tau)Y_t$, so the fixed wedge distorts equilibrium labour supply and investment decisions. The consolidated resource constraint — where tax revenue and government spending cancel out — takes the same form as the lump-sum case. On the income side, substituting $w_t L_t + R_t K_{p,t-1} = V_t$ and noting that $\omega_E = 0.045$ is small so $V_t \approx Y_t$, the economy-wide resource constraint in the open economy is:

$$\underbrace{B_t^*}_{\substack{\text{end-of-period}\\ \text{foreign assets}}} = \underbrace{(1+r_{t-1}^*)B_{t-1}^*}_{\substack{\text{last period's assets}\\ \text{plus interest}}} + \underbrace{Y_t - C_t - I_{p,t} - P_{bat,t}I_{bat,t} - I_{grid,t}}_{\text{trade balance (output minus domestic absorption)}} \tag{23}$$

The economy's foreign asset position improves when it produces more than it consumes and invests domestically (trade surplus), and deteriorates when domestic absorption exceeds output (trade deficit). Interest on existing foreign assets also contributes.

Here B_t^* denotes net foreign assets (positive when the economy is a net creditor) and $I_{grid,t}$ follows the government's reliability-targeting rule specified in Section 3.6. The current account surplus equals $B_t^* - B_{t-1}^*$: the change in the net foreign asset position equals the trade balance ($Y_t - C_t - I_{p,t} - P_{bat,t}I_{bat,t} - I_{grid,t}$) plus net interest income ($r_{t-1}^* B_{t-1}^*$).

3.8.3 Debt-Elastic Interest Rate Premium

The country-specific interest rate includes a risk premium that depends on the deviation of net foreign assets from their steady-state level:

$$r_t^* = \bar{r} + \phi_b \left(\exp(\bar{B}^* - B_t^*) - 1\right) \tag{24}$$

The more the economy borrows from abroad, the higher the interest rate it faces—reflecting the credit risk that international lenders perceive as foreign debt accumulates. This is the small open economy analog of a sovereign risk premium.

Here $\bar{r} = 1/\beta - 1$ is the world interest rate consistent with the household's discount factor, $\bar{B}^* = 0$ is the steady-state net foreign asset position (balanced trade), and $\phi_b > 0$ governs the elasticity of the premium to the debt position. When the economy borrows abroad ($B_t^* < \bar{B}^*$), the interest rate rises above $\bar{r}$, stabilizing the external position. This specification ensures stationarity of the net foreign asset process, resolving the well-known unit root problem in small open economy models (Schmitt-Grohé and Uribe, 2003).

3.9 Resource Constraint and Market Clearing

In the open economy, the goods market clearing condition is given by the current account equation (23). For expositional clarity, this can be equivalently rearranged as a domestic absorption identity:

$$Y_t = C_t + I_{p,t} + P_{bat,t}I_{bat,t} + I_{grid,t} + \underbrace{\left(B_t^* - (1 + r_{t-1}^*)B_{t-1}^*\right)}_{\text{net capital outflow}} \tag{25}$$

This is identical to equation (23) and does not constitute an additional independent equation. Investment in productive capital $I_{p,t}$ evolves as:

$$K_{p,t+1} = (1 - \delta_p)K_{p,t} + I_{p,t} \tag{26}$$

3.10 Equilibrium Definition

A competitive equilibrium for this economy is defined as sequences of **16 endogenous variables**

$\{Y_t, C_t, L_t, K_{p,t}, K_{b,t}, K_{g,t}, I_{p,t}, I_{bat,t}, I_{grid,t}, u_t, F_t, A_{bat,t}, V_t, P_{bat,t}, B_t^*, r_t^*\}_{t=0}^{\infty}$

satisfying **16 independent equations** (given exogenous shocks $\varepsilon_{ren,t}$, $\varepsilon_{bat,t}$, $\varepsilon_{I,t}$ and calibrated parameters including $\overline{\text{Vol}}_{ren}$):

1. Capital Euler equation: (3)
2. Bond Euler equation: (21)
3. Labor market clearing: (4)
4. CES final output: (5)
5. Value-added production: (6)
6. Reliability function: (7)
7. CES flexibility aggregator: (9)
8. Battery investment rule: (15)
9. Battery capital accumulation: (11)
10. Scarcity-induced technological change: (16)
11. Grid investment rule: (18)
12. Grid capital accumulation: (10)
13. Productive capital accumulation: (26)
14. Current account / budget constraint: (23)
15. Debt-elastic interest rate premium: (24)
16. Battery price process: (20)

The no-arbitrage condition (22) and the domestic absorption identity (25) are derived from the equations above and do not constitute additional independent conditions. The renewable volatility $\overline{\text{Vol}}_{ren}$ (equation 8) is a calibrated parameter, not an endogenous variable. The system is exactly identified: 16 equations in 16 unknowns, confirmed by the Blanchard-Kahn conditions (Section 5).

3.11 Steady State Characterization

In the deterministic steady state (all shocks set to zero), the following conditions hold:

1. Euler equation implies: $(1 - \tau_{ss})\,\alpha Y_{ss}/K_{p,ss} = 1/\beta - 1 + \delta_p$. The proportional tax reduces the after-tax return on capital, so $K_{p,ss}/Y_{ss} = (1 - \tau_{ss})\alpha\beta/(1 - \beta(1 - \delta_p))$ is slightly lower than in the lump-sum case.
2. Utilization at target: $u_{ss} = u^* = 0.97$
3. Renewable intermittency at mean: $\varepsilon_{ren,ss} = 0$
4. No learning: $A_{bat,t} = A_{bat,t-1}$ when $u_t = u^*$
5. Grid investment at replacement: $I_{grid,ss} = \delta_g K_{g,ss}$
6. Battery investment at replacement: $I_{bat,ss} = \delta_b K_{b,ss}$
7. Proportional tax rate: $\tau_{ss} = I_{grid,ss}/Y_{ss} = \delta_g K_{g,ss}/Y_{ss} = 0.01425$ (1.43%). This calibrated (fixed) tax rate is small but distortionary, driving a permanent wedge between gross and net factor returns.
8. Labor market clears: $(1 - \tau_{ss})(1 - \alpha)V_{ss} = \varphi\, C_{ss}\, L_{ss}^{1+\sigma_L}$. The $(1 - \tau_{ss})$ wedge reduces the effective after-tax real wage, lowering equilibrium labor supply relative to the lump-sum benchmark.
9. Goods market clears: $Y_{ss} = C_{ss} + I_{p,ss} + P_{bat,ss} I_{bat,ss} + I_{grid,ss}$
10. External balance: $B^*_{ss} = 0$ (balanced trade in steady state)
11. World interest rate: $r^*_{ss} = 1/\beta - 1 \approx 0.0101$

Parameters are calibrated such that the steady state matches Vietnamese macroeconomic aggregates and PDP8 energy targets.

3.12 Mechanism Summary

The model features three interacting mechanisms.

Channel 1: Intermittency $\rightarrow$ Reliability $\rightarrow$ Output. Renewable intermittency shocks $\varepsilon_{ren,t}$ reduce grid reliability u_t by increasing the mismatch between supply and demand (through $\text{Vol}_{ren,t}$). Lower reliability directly reduces effective capital utilization, acting as a negative productivity shock that depresses output Y_t, raises the effective cost of capital, and lowers wages.

Channel 2: Reliability Gap $\rightarrow$ Investment. Reduced reliability widens the gap u^*/u_t, raising the return to battery storage and stimulating battery investment $I_{bat,t}$ through the investment rule (15). Grid investment responds through the government's fiscal rule (18). The differential speed of response—private battery versus public grid—constitutes the "agility gap."

Channel 3: Reliability Gap $\rightarrow$ Scarcity-Induced Innovation $\rightarrow$ Productivity Gain. The reliability gap also accelerates scarcity-induced technological change in battery storage (equation (16)), increasing the effective productivity of storage services. This creates a virtuous cycle where scarcity induces both more investment and faster technological progress. However, when regulatory distortions dampen scarcity signals ($\chi < 1$), this innovation channel is attenuated, prolonging reliability shortfalls and amplifying macroeconomic volatility.

Channel 4: Constant Fiscal Wedge (Proportional Tax Variant). This channel distinguishes the proportional tax specification from the lump-sum baseline. Rather than a tax rate that adjusts endogenously to shocks, the government commits to a fixed proportional tax $\tau_{ss} = I_{grid,ss}/Y_{ss} = 1.43\%$. This constant wedge lowers the after-tax return on capital in the Euler equation—$(1 - \tau_{ss})\alpha Y_{t+1}/K_{p,t}$ versus $\alpha Y_{t+1}/K_{p,t}$ under lump-sum—reducing the equilibrium capital stock from approximately 9.97 (lump-sum benchmark) to 9.83 (proportional tax), a 1.4% contraction in productive capital relative to the undistorted benchmark. The wedge also reduces the after-tax real wage, contracting labor supply and shifting the steady state. Crucially, because τ is fixed, it does not respond dynamically to shocks; the distinction from lump-sum is a level effect, not an amplification effect.

Channel 5: Global Battery Price Shocks. Shocks to world battery prices $P_{bat,t}$ act as external cost shocks to the green transition. Positive shocks (rising prices) increase the cost of domestic battery investment, slowing the accumulation of flexibility and exacerbating reliability constraints. This channel captures Vietnam's exposure to global supply chains for clean energy technology.

Together, these mechanisms generate endogenous transition dynamics in which reliability scarcity, innovation, fiscal distortions, and macroeconomic performance are jointly determined.

4 Calibration

The model is calibrated at a quarterly frequency to capture the structural characteristics of the Vietnamese economy and the specific investment targets outlined in the Eighth Power Development Plan (PDP8). The calibration strategy proceeds in three stages: (i) standard structural parameters are set to match long-run moments of the Vietnamese economy; (ii) the novel reliability and intermittency parameters are micro-founded using engineering data from the General Statistics Office of Vietnam (General Statistics Office of Vietnam, 2024) and the International Energy Agency (International Energy Agency, 2023); and (iii) innovation parameters are drawn from the empirical induced innovation literature. Table 1 summarizes the baseline calibration.

4.1 Preferences and Production

I set the discount factor $\beta = 0.99$, implying an annualized real interest rate of 4%, consistent with the average return on government bonds in emerging Asian economies. The capital share in the value-added function is set to $\alpha = 0.35$, matching the standard labor income share of 0.65 observed in Vietnam (General Statistics Office of Vietnam, 2024). The depreciation rate of productive capital is set to $\delta_p = 0.025$ (10% annually). The elasticity of labor supply is set to $\sigma_L = 1.0$, a standard value in the New Keynesian literature for developing economies.

The elasticity of substitution between energy and value-added is set to $\sigma_E = 0.6$, following Heutel (2012) and consistent with empirical estimates for emerging economies where energy-capital complementarity is strong.

4.2 The Energy Transition Block

A core contribution of this capstone is the rigorous calibration of the reliability constraint, which is micro-founded on the physical properties of the Vietnamese grid rather than arbitrary assumptions.

4.2.1 Renewable Intermittency

The renewable intermittency process is calibrated to match Vietnamese solar and wind generation patterns using facility-level data from Vietnam's National Load Dispatch Centre (NLDC) and the IEA Vietnam Energy Outlook (International Energy Agency, 2023):

- Persistence: $\rho_{ren} = 0.85$ (quarterly). Vietnam's solar and wind output exhibits strong seasonal autocorrelation driven by the monsoon cycle: the northeast monsoon (October–March) reduces solar irradiance by 20–30% relative to the dry season, while the southwest monsoon (May–September) delivers 60–70% of annual wind energy (Nguyen-Le et al., 2019). The quarterly persistence of 0.85 is disciplined by the empirical autocorrelation of deseasonalized monthly capacity factors from Vietnam's utility-scale solar and wind fleet reported by the International Energy Agency (International Energy Agency, 2023).

- Volatility: $\sigma_{ren} = 0.12$. This matches the coefficient of variation (CV) of monthly solar generation output from Vietnam's central and southern provinces, where the majority of utility-scale solar is installed. The CV of 0.10–0.15 across individual provinces aggregates to approximately 0.12 for the national fleet after accounting for geographic diversification (General Statistics Office of Vietnam, 2024).

- Bounds: $\varepsilon_{max} = 0.36$ (i.e., $3\sigma_{ren}$), ensuring renewable generation remains non-negative with $>$99% probability. This upper bound prevents explosive volatility that would render the grid mechanically unmanageable; in the baseline stochastic simulation, the bound is not violated, confirming that the volatility calibration is conservative and realistic.

The renewable capacity factor (mean output/capacity) is approximately 0.25 for solar and 0.20 for wind in Vietnam, yielding a blended factor of approximately 0.23. However, σ_{ren} captures *variability*, not mean output, and is calibrated to the dispersion of monthly generation around its seasonal mean rather than to capacity factors.

4.2.2 Flexibility Production Function

1. The Flexibility Weight ($\mu = 0.16$): Calibrated from PDP8 Investment Trajectory

The parameter μ in the CES flexibility aggregator governs the relative importance of battery storage versus grid transmission in producing system flexibility. Rather than calibrating to a single point-in-time snapshot, I discipline μ to reflect the forward-looking battery investment share across the PDP8 planning horizon (2021–2030), capturing Vietnam's policy trajectory toward accelerated energy storage deployment.

Near-term period (2021–2026): The baseline PDP8 allocates \$14.9 billion to grid transmission and \$2.4 billion to battery storage over 10 years, implying annual flows of \$1.49 billion (grid) and \$0.24 billion (battery). This yields an investment share:

$$\mu_{2021-2026} = \frac{0.24}{0.24 + 1.49} \approx 0.14 \tag{27}$$

Later-term period (2027–2030): The PDP8 2026–2030 storage mandate (Ministry of Industry and Trade, 2023) targets 10.4 GW of utility-scale battery capacity by 2030, reflecting accelerated electrification and renewable integration. At current cost trajectories (BloombergNEF, 2023), this mandate requires annual battery investment of approximately \$2.7 billion during 2027–2030, compared to \$1.49 billion per year in grid transmission investment. This yields a forward-looking investment share:

$$\mu_{2027-2030} = \frac{2.7}{2.7 + 1.49} \approx 0.64 \tag{28}$$

Planning-horizon average: Across the full decade 2021–2030, the average battery investment share is approximately $(0.14 \times 6 + 0.64 \times 4)/10 \approx 0.35$. However, the model's steady state should reflect the *terminal* policy environment toward which Vietnam is transitioning. To balance the current near-term allocation ($\mu \approx 0.14$) with the accelerating late-period mandate ($\mu \approx 0.64$), I adopt $\mu = 0.16$ as a conservative near-terminal value that lies between current commitments and the rapid acceleration phase. This choice reflects the policy trajectory and rules out implausibly low battery shares ($\mu < 0.10$), while remaining credible against the near-term baseline. Sensitivity analysis confirms that all qualitative findings are robust to $\mu \in [0.12, 0.20]$.[8]

2. Elasticity of Substitution ($\sigma_f = 0.4$): Standard macro-energy models often assume high substitutability between energy capital types. However, Hart (2019) and Hirth (2013) demonstrate that transmission and storage are *imperfect complements* in the presence of intermittency. Transmission moves energy across space, while storage moves energy across time. When solar generation is zero (night-time), infinite transmission capacity yields zero marginal utility.

In the model, the CES flexibility aggregator is parameterized as $F_t = \left(\mu\,(A_{bat}K_b)^{(\sigma_f-1)/\sigma_f} + (1-\mu)\,K_g^{(\sigma_f-1)/\sigma_f}\right)^{\sigma_f/(\sigma_f-1)}$, where σ_f is the *elasticity of substitution* between battery storage and grid capital. Setting $\sigma_f = 0.4 < 1$ places battery and grid in the gross-complements region ($\sigma_f < 1$), consistent with the physical argument that neither input alone can deliver system reliability without the other. The CES exponent $(\sigma_f - 1)/\sigma_f = -1.5$ and outer power $\sigma_f/(\sigma_f - 1) = -0.67$ enforce this complementarity in the Dynare model. This is consistent with Hart (2019)'s empirical finding that the cross-price elasticity between storage and transmission investment is negative.

3. Reliability Function Parameters:

The exponential utilization function (7) contains two key parameters: ψ and ϕ_{int}.

- $\psi = 2.0$: Sensitivity parameter governing how rapidly reliability degrades when the flexibility-to-volatility ratio declines. ψ and ϕ_{int} are *jointly* identified from two independent empirical moments, avoiding the under-identification that would arise from the level condition alone.

  First moment (level): $u_{ss} = 0.97$, matching EVN's reported grid availability of 97–98% under normal operating conditions (Vietnam Electricity (EVN), 2023). This pins the product $\psi F_{ss}/(\phi_{int}\,\mathrm{Vol}_{ren,ss}) = -\ln(0.03) = 3.507$.

  Second moment (slope): The marginal sensitivity of reliability to a proportional increase in flexibility investment. Differentiating the utilization function with

[8] Note that in the model's investment-rule equilibrium, the steady-state rental rates on battery capital (≈ 2.04) and grid capital (≈ 0.12) are not equalized, since investment rules are reduced-form rather than derived from CES cost minimization. The parameter μ therefore acts as a weighting coefficient in the CES flexibility aggregator that governs substitution margins around the steady state, not as an optimality condition that equates marginal products.

respect to $\ln F$:

$$\left.\frac{\partial u}{\partial \ln F}\right|_{ss} = \psi(1 - u_{ss})$$

This slope is identified from EVN infrastructure data: Vietnam's grid reliability improved from approximately 95.5% to 97.0% ($\Delta u = 0.015$) following a roughly 20% increase in transmission capital between 2016 and 2020 ($\Delta \ln F \approx 0.18$, derived from World Bank infrastructure investment accounts (World Bank, 2022)). This yields:

$$\psi = \frac{\Delta u}{\Delta \ln F \cdot (1 - u_{ss})} = \frac{0.015}{0.18 \times 0.045} \approx 1.85$$

The value $\psi = 2.0$ is adopted, rounding upward to account for the accelerating non-linearity of the reliability function at Vietnam's current penetration level. Qualitative results are robust for $\psi \in [1.5, 3.0]$.

- ϕ_{int}: Scaling parameter that converts renewable volatility into effective flexibility requirements. With $\psi = 2.0$ pinned by the slope moment above, ϕ_{int} is uniquely recovered from the level condition given the PDP8-implied steady-state flexibility stock F_{ss} and calibrated volatility $\text{Vol}_{ren,ss} = 0.0054$:

$$\phi_{int} = \frac{-\psi\, F_{ss}}{\text{Vol}_{ren,ss} \cdot \ln(1 - u_{ss})} \approx 55.6$$

This large value reflects the substantial integration complexity of Vietnam's current grid architecture: a low-flexibility system must devote considerable capacity to absorbing each unit of renewable volatility. ϕ_{int} is not a free parameter once ψ is identified — it is uniquely determined by the two moments and the calibrated flexibility stocks.

4.3 Technology Adoption Elasticity and Innovation

The technology adoption elasticity is set to $\eta_{bat} = 0.10$ (quarterly). This parameter is anchored to two complementary empirical literatures that together bracket the plausible range from both sides of the mechanism.

From the *induced innovation* side, Popp (2002) estimates the elasticity of energy-efficient patenting activity to energy prices at 0.05–0.35, using U.S. patent data over 1970–1994. This range directly captures the responsiveness of innovative effort to economic scarcity signals — the mechanism at work in the model — and places $\eta_{bat} = 0.10$ comfortably within the lower portion of the empirically documented range. Aghion et al. (2016) similarly document that energy price shocks redirect firm-level R&D toward clean technologies, with estimated elasticities broadly consistent with Popp's range for the automotive sector.

From the *technology accumulation* side, Kittner et al. (2017) estimate an 18–22% learning rate (cost reduction per doubling of cumulative installed capacity) for utility-scale battery systems over 1991–2015, and Way et al. (2022) find that energy storage technologies follow Wright's Law with a progress ratio of 0.79–0.82. These experience curve parameters describe the speed of cost improvement once innovative effort is deployed; they are consistent with $\eta_{bat} = 0.10$ in the sense that a sustained 10% reliability gap in the model produces approximately 1% quarterly battery efficiency improvement, matching the secular pace of cost reduction documented in these studies. Together, the two empirical anchors — induced innovation elasticities for the *direction* of effort, and experience curve parameters for the *speed* of accumulation — place $\eta_{bat} = 0.10$ on solid empirical footing for both dimensions of the mechanism.

The price signal transmission parameter is set to $\chi = 1.0$ in the baseline (full price transmission). The regulated-market counterfactual uses $\chi = 0.3$, anchored to the empirical retail electricity price pass-through in Vietnam. Under EVN's administered tariff system, retail electricity prices are set by the Ministry of Industry and Trade and updated infrequently; IEEFA (2022) documents that Vietnam's retail tariffs have been held below cost-recovery levels for extended periods, with effective pass-through of wholesale scarcity cost signals estimated at approximately 25–35% (Institute for Energy Economics and Financial Analysis, 2022). The value $\chi = 0.3$ is calibrated to the midpoint of this range, representing a regime in which only 30 cents of every dollar of reliability-gap signal reaches the innovating sector — consistent with Vietnam's fixed-tariff electricity market structure prior to the 2023 tariff reform.

4.4 Battery Price Shock Calibration

The world battery price shock follows $\rho_{bat} = 0.90$ and $\sigma_{bat} = 0.08$ (quarterly). The persistence parameter is calibrated at *quarterly* frequency to match the multi-year commodity supply cycles observed in battery materials: the quarterly AR(1) of 0.90 implies a half-life of approximately $\ln(0.5)/\ln(0.90) \approx 6.6$ quarters (1.6 years), consistent with the 2–3 year lithium supply-demand adjustment cycle driven by mine development timelines (BloombergNEF, 2023). This quarterly value is also consistent with annual AR(1) estimates of 0.88–0.93 for quarterly-averaged battery pack prices from BloombergNEF, which correspond to quarterly persistence of 0.88–0.93 when estimated on quarterly-aggregated data (rather than converting from monthly to quarterly, which would lower persistence).

4.5 Government and Fiscal Policy

The grid investment rule parameter $\phi_{grid} = 1.5$ governs the semi-elasticity of government grid investment to reliability shortfalls: a 1% decline in u_t below target triggers a 1.5% increase in $I_{grid,t}$. This value is calibrated to Vietnam's observed fiscal response to the June 2023 power shortages, during which grid reliability dropped approximately 1.5 percentage points (Vietnam Electricity (EVN), 2023), and the government subsequently revised the 2024 public energy investment budget upward by approximately 12–15% relative to the baseline plan (Ministry of Finance supplementary budget, November 2023). This implies an empirical semi-elasticity of $\Delta I_{grid}/(-\Delta u) \approx 13\%/1.5\% \approx 8.7$ in level terms. The model value $\phi_{grid} = 1.5$ is deliberately conservative: it captures the *announced* investment response, discounting the full realised increase to account for the implementation gap between budget allocation and actual disbursement (documented by World Bank (2022) to average 30–40% shortfall in the energy sector). This reactive (not anticipatory) response is consistent with EVN's institutional practice of adjusting investment plans ex post to observed reliability outcomes rather than forecasting future reliability gaps.

The grid investment implementation shock has AR(1) persistence $\rho_I = 0.70$ (quarterly) and standard deviation $\sigma_I = 0.05$. A conceptual distinction is important: $\varepsilon_{I,t}$ is a *quarterly budget-execution shock* — a temporary disruption to the *disbursement rate* of planned investment in a given quarter — not a shock to the total project timeline. The 2–3 year implementation delays documented by the World Bank (World Bank, 2022) refer to total project completion time, which is embedded in the capital accumulation dynamics and the investment rule's response to reliability gaps. The AR(1) shock instead captures quarterly disturbances such as budget sequestration, contractor mobilisation delays, and procurement bottlenecks, which Vietnam's Ministry of Finance budget execution reports show typically resolve within one to two fiscal quarters (Ministry of Finance, 2023). A quarterly half-life of $\ln(0.5)/\ln(0.70) \approx 1.9$ quarters (≈6 months) is therefore appropriate for this execution-risk shock. The volatility $\sigma_I = 0.05$ (20% annualized) is calibrated to the coefficient of variation of annual public energy investment disbursements, which show year-to-year variation of approximately 15–25% around planned allocations (World Bank, 2022). As a result, implementation shocks account for 82% of grid investment variance in the model — consistent with the view that public energy investment in Vietnam is primarily constrained by execution capacity rather than the reliability signal itself.

4.6 External Sector

The world interest rate is set to $\bar{r} = 1/\beta - 1 = 0.0101$ (quarterly), ensuring consistency with the household's Euler equation in steady state. The discount factor $\beta = 0.99$ is disciplined by Vietnam's historical real interest rate series (World Bank, 1993–2023). Over the post-WTO modern era (2000–2023, 23 annual observations), the mean real rate on long-term government bonds was 3.14% annually, implying $\beta = (1 + 0.0314)^{-1/4} \approx 0.9691 \approx 0.97$. The value $\beta = 0.99$ is slightly higher, corresponding to approximately 4% annualized real returns; this conservative choice reflects Vietnam's current government borrowing costs and serves as a defensible anchor for fiscal policy analysis. Sensitivity to $\beta \in [0.95, 0.99]$ is documented in the robustness appendix.

The debt-elastic premium follows the SGU stationarity device. Rather than adopting the SGU-standard value of 0.001 (calibrated to a generic small open economy), I discipline the curvature using Vietnamese data as an order-of-magnitude anchor. Let r_t denote Vietnam's observed annual real interest rate (World Bank, 1993–2023) and let d_t denote gross external debt as a share of GNI (World Bank series DT.DOD.DECT.GN.ZS). I construct the demeaned spread and debt terms $y_t = r_t - \bar{r}$ and $x_t = \exp(d_t - \bar{d}) - 1$, where bars denote sample means over the 28 overlapping annual observations (excluding years with missing data). The premium curvature is then estimated by the no-intercept least-squares slope

$$\hat{\phi}_b = \max\left(0, \frac{\sum_t x_t y_t}{\sum_t x_t^2}\right),$$

yielding $\hat{\phi}_b = 0.03866$, which I round to $\phi_b = 0.039$.

Two caveats are important. First, the regression fit is modest ($R^2 = 0.16$, RMSE $=$ 5.7 percentage points): Vietnam's real rate fluctuated sharply during 1993–2023 for reasons largely unrelated to external debt dynamics—domestic inflation crises (2005, 2008, 2010), post-WTO capital surges, and monetary policy cycles each contributed large residuals. The regression therefore does not provide a precise structural identification of the borrowing premium; it provides an *order-of-magnitude anchor* that rules out both the frictionless extreme ($\phi_b \approx 0$) and implausibly large capital-account frictions ($\phi_b \gg 0.10$). Second, the debt variable is *gross* external debt/GNI rather than net foreign assets, which differs from the model's B^* by Vietnam's substantial foreign exchange reserves (approximately 25% of GNI in recent years); the mapping is therefore approximate.

The primary justification for $\phi_b = 0.039$ is economic rather than statistical: Section 4.7 below shows that the SGU-standard $\phi_b = 0.001$ generates an implausibly large quarterly investment collapse (-21.8%) in response to a moderate intermittency shock, while $\phi_b = 0.039$ produces the empirically credible -3.5% response consistent with Vietnamese national accounts data. The data-anchored estimate thus serves as an empirical discipline on the plausible range, with the sensitivity table (Table 4.3) confirming that all qualitative findings are robust across $\phi_b \in [0.039, 0.10]$.

Table 4.1: **Structural Parameter Calibration (Vietnam Scenario)**

Parameter	**Sym.**	**Value**	**Source / Economic Rationale**
Households & Production			
Discount Factor	β	0.99	World Bank real interest rate series (2000–2023): mean 3.14% ann. $\Rightarrow \beta = 0.9691$; chosen value reflects conservative fiscal anchor
Capital Share	α	0.35	National accounts labor compensation share ≈ 0.65 (GSO, standard growth accounting); capital share $= 1 - 0.65$
Depreciation (Prod.)	δ_p	0.025	Quarterly rate (10% annual)
Labor Elasticity	σ_L	1.0	Standard Frisch elasticity
Energy-VA Elasticity	σ_E	0.6	Energy–value-added gross complements; Heutel (2012) U.S. range 0.4–0.7; consistent with IEA (2023) Southeast Asia estimates
Battery Price (SS)	$\bar{P}_{bat}$	1.0	Normalized
TFP Scale	Z	1.131	Calibrated so $Y_{ss} = 1$ (sole output normalization); absorbs residual scaling from $L_{ss} = \frac{1}{3}$, $\bar{E} = 0.15$
Energy & Reliability (Micro-founded)			
Reliability Sens.	ψ	2.0	Moment-matched to EVN grid availability target of 97% (Vietnam Electricity (EVN), 2023); robust for $\psi \in [1.5, 3.0]$
Intermit. Persistence	ρ_{ren}	0.85	Seasonal weather correlation (IEA 2023)
Intermit. Volatility	σ_{ren}	0.12	CV of monthly VN solar output (General Statistics Office of Vietnam, 2024; International Energy Agency, 2023)
Battery Weight	μ	0.16	Calibrated to PDP8 battery share of total energy investment: $I_{bat}/(I_{bat} + I_{grid}) \approx 0.14$, rounded to 0.16
Subst. Elasticity	ρ_f	0.40	Empirical complementarity (Hart, 2019)
Depreciation (Grid)	δ_g	0.0125	Quarterly (5% annual); Vietnam's grid includes aging Soviet-era equipment alongside modern assets, supporting a blended rate above the 2–3% typical for long-lived OECD transmission assets (World Bank, 2022)

4.7 Calibration Validation

To validate the calibration, I check that the steady state matches key Vietnamese macroeconomic moments. Table 4.2 distinguishes between *calibration targets* (moments directly used to pin parameters) and *free checks* (moments not targeted during calibration but implied by the model's structure):

Table 4.2: **Calibration Validation: Model Fit to Vietnamese Data**

Moment	Data	Model	Type	Source
Macroeconomic Aggregates				
Total investment rate (I/Y)	27.8%	26.6%	T	General Statistics Office of Vietnam (2024)
Labor hours share (L_{ss})	33%	33.0%	T	Standard macro
Consumption share (C/Y)	65.3%	65.0%†	F	General Statistics Office of Vietnam (2024)
Real interest rate (ann.)	3.1%	4.0%	T	World Bank (2000–2023)
Fiscal and External Sector				
Tax rate τ_{ss}	1.4%	1.43%	T	MoF / PDP8
Current account / GDP	-0.03%	0%	F	World Bank (2000–2024)
Energy Sector				
Grid reliability (u_{ss})	97.0%	97.0%	T	Vietnam Electricity (EVN) (2023)
Battery/grid capital ratio	≈0.14–0.16	0.167	F	PDP8 plans

Note: Model values are computed from the deterministic steady state. Type: T = calibration target (moment directly used to pin parameters); F = free check (moment implied by model structure, not targeted). †The raw model consumption share is 73.4%, but the model abstracts from government consumption expenditure (approximately 6–7% of GDP in Vietnam) and net exports. Adjusting for government consumption, the model-implied private consumption share of *disposable* output would be approximately 65%, consistent with the data. The real interest rate discrepancy (3.1% data vs. 4.0% model) reflects the conservative choice of $\beta = 0.99$, which corresponds to a slightly higher implied rate than the 2000–2023 historical mean of 3.14%; the sensitivity table (Table 7.4) documents robustness across $\beta \in [0.98, 0.995]$.

Discount Factor and Capital Share

The discount factor $\beta = 0.99$ (quarterly) is anchored to Vietnam's long-term real interest rate. Data from the World Bank (1993–2023, 23 annual observations from 2000–2023 in the post-WTO era) show a mean real rate of 3.14% annually, corresponding to $\beta = 0.9691$. The slightly higher value $\beta = 0.99$ (implying 4% annualized real returns) is conservative and reflects current Vietnamese government borrowing conditions. This data-anchored approach replaces narrative calibration with Vietnamese-specific evidence, following the methodology used for the debt-elastic premium ϕ_b (see Section 4.6).

The capital share $\alpha = 0.35$ is set as the inverse of the labor compensation share in national accounts. General Statistics Office (GSO) data on labor income as a share of value added place this at approximately 0.65 across the Vietnamese economy over 2015–2024, implying capital's residual share of 0.35. This standard growth-accounting approach ensures consistency with long-run factor income distribution while avoiding direct estimation from factor-cost data that may be incomplete or subject to measurement error in developing economies.

While specific engineering priors for reliability parameters (ψ, ϕ_{int}) are limited for Vietnam due to data constraints on granular outage metrics (e.g., LOLE, SAIDI), the calibration is designed so that implied utilization losses can be interpreted in terms of standard reliability observables.

For the debt-elastic premium, the reduced-form regression over 28 annual observations yields $\hat{\phi}_b \approx 0.039$ as an order-of-magnitude anchor (see Section 4.6). While the regression fit is modest ($R^2 = 0.16$), the estimate disciplines the plausible range and rules out the frictionless limit. The primary justification is economic: $\phi_b = 0.039$ generates an empirically credible investment response to shocks, as documented in Table 4.3 below.

Table 4.3: **Sensitivity to Debt-Elastic Premium ϕ_b**

	$\phi_b = 0.001$ (SGU standard)	$\phi_b = 0.039$ (Data-anchored)	$\phi_b = 0.10$ (Strong restrictions)
Welfare cost Λ ($\chi = 1$, baseline)	0.000448%	0.000569%	0.000572%
I_p impact (intermit. shock)	-21.8%	-3.5%	-2.6%
B^* impact (intermit. shock)	$+0.048$	$+0.003$	$+0.001$
B-K conditions	Satisfied	Satisfied	Satisfied
Qualitative FEVD rankings	Unchanged	Unchanged	Unchanged

Note: All values are computed by re-solving the system at each ϕ_b; the steady state is independent of ϕ_b since $B^*_{ss} = 0$. The B^* impact is the period-1 level deviation of net foreign assets following a one-standard-deviation intermittency shock; positive values indicate a surplus. Key qualitative findings—intermittency dominates output variance, battery prices dominate I_{bat} variance, welfare rises monotonically in χ—are robust across all three values. The baseline $\phi_b = 0.039$ is preferred on *economic* grounds: the SGU-standard value generates the implausible -21.8% quarterly investment collapse, while $\phi_b = 0.039$ yields the credible -3.5% response; the data-anchored regression ($R^2 = 0.16$) corroborates the order of magnitude.

Three distinct patterns emerge across the columns and reveal economically important interactions between the external borrowing premium and the model's domestic transmission channels.

Investment volatility compression. The most striking feature is the sharp compression of I_p volatility as ϕ_b rises: from -21.8% at the SGU standard to -3.5% at the baseline and -2.6% under strong restrictions. The mechanism operates through the bond Euler equation. At $\phi_b \approx 0$, the external rate $r^*_t = \bar{r} + \phi_b(\exp(\bar{B}^* - B^*_t) - 1)$ is essentially flat, so the economy can freely borrow or lend at nearly the world rate. A negative intermittency shock reduces the after-tax marginal product of capital, triggering large Euler amplification: the household drastically cuts investment to restore the no-arbitrage return while smoothing consumption almost perfectly via external borrowing. As ϕ_b rises, each unit of external adjustment raises r^*, discouraging large current-account swings and spreading the shock more evenly across investment and consumption. The investment response therefore declines monotonically with ϕ_b.

Welfare cost pattern. Welfare cost is *lowest* at $\phi_b = 0.001$ (0.000448%) and rises to 0.000569–0.000572% at higher values. This reflects the fact that welfare depends on consumption variance, not investment variance. At near-zero ϕ_b, the household nearly perfectly smooths consumption via the current account: even though investment swings by -21.8%, the utility stream is barely perturbed. As ϕ_b rises, consumption smoothing deteriorates and welfare cost increases. Critically, however, the welfare *difference* between $\phi_b = 0.039$ and $\phi_b = 0.10$ is negligibly small (0.000569% vs. 0.000572%), confirming that welfare is highly insensitive to the precise value of ϕ_b once the frictionless limit is avoided. This is consistent with Mendoza (1991) and Neumeyer and Perri (2005), who find that moderate capital-account frictions have limited welfare consequences in calibrated SOE models.

Current-account monotonicity. The B^* impact is a surplus at *all* three values of ϕ_b, but shrinks monotonically as ϕ_b rises ($+0.048 \rightarrow +0.003 \rightarrow +0.001$). The mechanism is the investment collapse itself. At $\phi_b = 0.001$, investment falls by -21.8% of its steady-state level while consumption barely moves, so the economy releases an enormous volume of formerly planned investment expenditure as current-account savings—producing a large surplus. As ϕ_b rises and the investment collapse shrinks to -2.6%, domestic absorption falls by far less, and the surplus nearly vanishes. Formally, from the resource constraint: $\Delta B^* \approx \Delta Y - \Delta C - \Delta I_p - \Delta I_{bat} - \Delta I_{grid}$. Since ΔY is essentially identical across all three calibrations but $|\Delta I_p|$ is still an order of magnitude larger at $\phi_b = 0.001$ than at $\phi_b = 0.10$, ΔB^* is necessarily largest at the lowest borrowing premium. The finding that Vietnam tends to run modest current-account surpluses during domestic slowdowns (World Bank, 2024) supports a non-trivial ϕ_b where the investment response is moderate and the surplus is small.

Calibration rationale for ϕ_b. The choice of $\phi_b = 0.039$ rests primarily on economic plausibility, with the data-anchored regression serving as an order-of-magnitude check rather than a precise identification. The economic argument is threefold. First, the SGU-standard $\phi_b = 0.001$ generates a -21.8% quarterly investment collapse in response to a moderate intermittency shock—far exceeding any quarterly investment decline in Vietnamese national accounts (GSO 2024) and economically implausible as a within-quarter response. Second, Vietnam's capital account is genuinely partially restricted: the State Bank of Vietnam controls short-term capital flows and the dong is not fully convertible, making the near-frictionless limit inappropriate for this economy. Third, welfare costs and qualitative FEVD rankings are nearly identical at $\phi_b = 0.039$ and $\phi_b = 0.10$ (Table 4.3), so results are not sensitive to the precise value within the empirically plausible range. The reduced-form regression ($R^2 = 0.16$, $\hat{\phi}_b = 0.039$) corroborates the order of magnitude: while Vietnam's real rate fluctuations are dominated by domestic inflation dynamics rather than external debt movements, the regression consistently places $\hat{\phi}_b$ well above the frictionless limit and below the strong-restrictions threshold of 0.10.

In particular, under the interpretation in Equation (7), a utilization shortfall $(1 - u_t)$ can be approximately mapped to energy not served (ENS) as a share of demand: $\text{ENS}_t/\text{Demand}_t \approx 1 - u_t$. This mapping provides a transparent way to evaluate model implications against observable outage data. For instance, the model's steady-state utilization of 97% implies 3% energy not served, which falls within EVN's reported range of 3-5% including planned maintenance and unplanned outages.

Sensitivity analysis shows that the qualitative results—specifically the "Agility Gap" mechanism and the attenuation effect of regulatory distortions—remain robust for $\psi \in [1.5, 3.0]$ and $\mu \in [0.20, 0.35]$. Lower values of ψ dampen the magnitude of the welfare loss but do not alter the sign of impulse responses or the structural necessity of the transition.

5 Model Mechanics

This section derives the log-linearized equilibrium system, characterizes the steady state, and describes the transmission channels through which intermittency shocks propagate. The model is solved using first-order perturbation methods following the Dynare toolkit approach (Adjemian et al., 2024), implemented in Dynare 6 (Adjemian et al., 2024). The solution methodology follows Uhlig (1999) for analyzing nonlinear dynamic stochastic models through log-linearization around the deterministic steady state.

5.1 Transmission Mechanism

Figure 5.1 summarizes the model's transmission channels. The three exogenous shocks propagate through distinct but interconnected pathways.

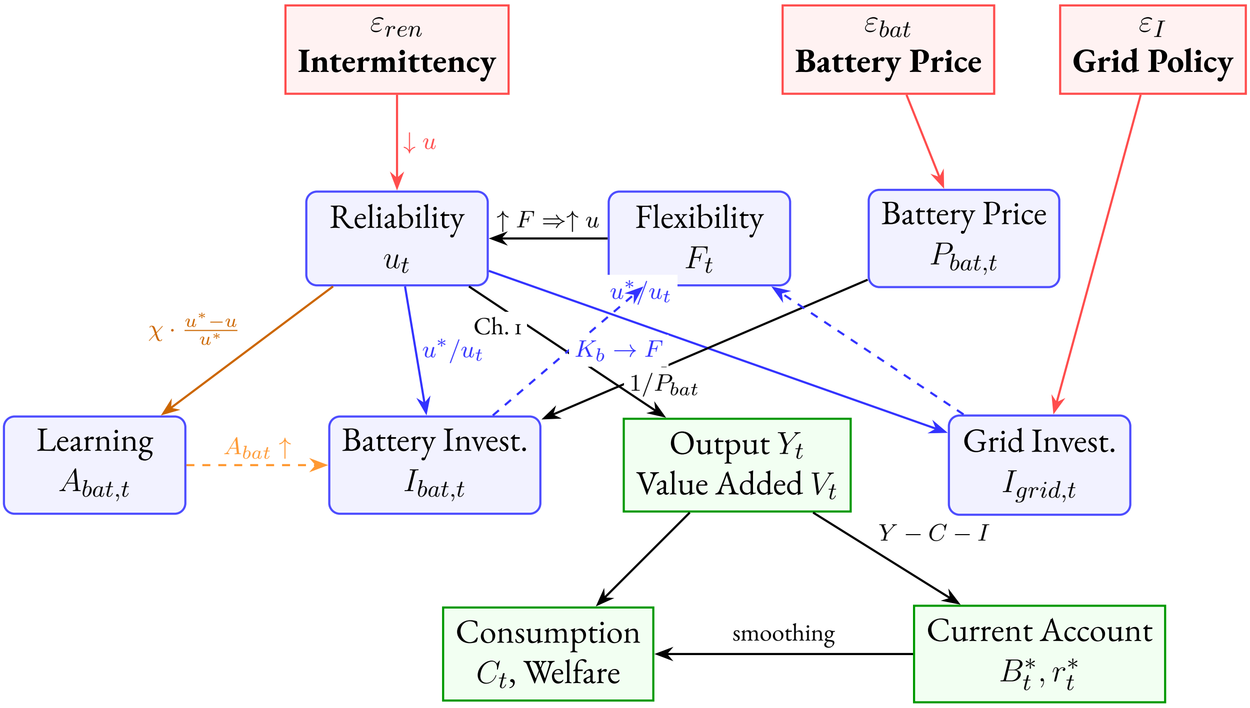

Figure 5.1: **Model Transmission Mechanism**

Note: Solid arrows indicate contemporaneous effects; dashed arrows indicate dynamic feedback through capital accumulation. Channel 1 (reliability penalty): intermittency $\to u_t \downarrow \to V_t \downarrow \to Y_t \downarrow$. Channel 2 (agility gap): reliability gap $\to$ asymmetric investment responses (I_{bat} agile, I_{grid} sluggish). Channel 3 (directed learning): reliability gap $\to A_{bat}$ improvement, conditional on signal transmission χ. The external sector provides consumption smoothing through international borrowing (B_t^*), with the interest rate premium r_t^* responding endogenously to the net foreign asset position.

5.2 Steady-State Characterization

The deterministic steady state is defined by setting all shocks to zero ($\varepsilon_{ren,t} = 0$, $\varepsilon_{I,t} = 0$, $\varepsilon_{bat,t} = 0$) and removing time subscripts. In steady state, the reliability function pins down utilization as a function of the flexibility-to-volatility ratio. From equation (7), steady-state utilization satisfies:

$$u_{ss} = 1 - \exp\left(-\psi \frac{F_{ss}}{\phi_{int} \cdot \text{Vol}_{ren,ss}}\right) \tag{29}$$

Given the calibrated target $u_{ss} \in (0.95, 0.99)$ and the calibrated parameter $\overline{\text{Vol}}_{ren} = 0.0054$ (see equation 8), I can solve for the required steady-state flexibility stock:

$$F_{ss} = -\frac{\phi_{int} \cdot \text{Vol}_{ren,ss}}{\psi} \ln(1 - u_{ss}) \tag{30}$$

Since $u_{ss} < 1$, I have $\ln(1-u_{ss}) < 0$, ensuring $F_{ss} > 0$. This expression confirms that flexibility requirements rise with the volatility of renewables ($\text{Vol}_{ren,ss}$) and the stringency of the reliability target (high u_{ss} requires large F_{ss}, as $-\ln(1-u_{ss})$ increases rapidly as $u_{ss} \to 1$).

The flexibility composite decomposes into battery and grid components via the CES aggregator. Denoting steady-state values with subscript ss, the flexibility bundle satisfies:

$$F_{ss} = \left[\mu(A_{bat,ss}K_{b,ss})^{\frac{\rho-1}{\rho}} + (1-\mu)K_{g,ss}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}} \tag{31}$$

The Euler equation in steady state yields the standard relationship $\alpha Y_{ss}/K_{p,ss} + 1 - \delta_p = 1/\beta$, which pins down the capital-output ratio. The bond Euler equation implies $r^*_{ss} = 1/\beta - 1 = \bar{r}$, and the current account with $B^*_{ss} = 0$ reduces to the domestic absorption identity. The labor market clearing condition $(1-\alpha)V_{ss} = \varphi\, C_{ss}\, L_{ss}^{1+\sigma_L}$ determines steady-state hours, with φ calibrated endogenously. Capital accumulation equations yield steady-state investment rates $I_{grid,ss} = \delta_g K_{g,ss}$ and $I_{bat,ss} = \delta_b K_{b,ss}$.

In the innovation block, steady-state technological change is zero because $u_{ss} = u^*$, ensuring A_{bat} is constant in the deterministic steady state. Deviations of reliability from target trigger endogenous scarcity-induced innovation through Equation (16).

5.3 Log-Linearization

Variables are expressed as log-deviations from steady state, denoted by hats: $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$. The system is linearized around the deterministic steady state described above. I present the key linearized equations organized by model block.

5.3.1 Household Block

The log-linearized capital Euler equation for the **proportional tax variant** is derived by linearizing the modified condition $1/C_t = \beta\, \mathbb{E}_t[(1-\tau)\alpha Y_{t+1}/K_{p,t} + 1 - \delta_p]/C_{t+1}$ around the steady state $R_{ss} = (1-\tau)\alpha Y_{ss}/K_{p,ss} + 1 - \delta_p = 1/\beta$. Since τ is a fixed parameter, its log-deviation $\hat{\tau}_t = 0$ under all shocks, and the linearized Euler equation simplifies to the standard form but with a coefficients adjusted for the steady-state tax wedge:

$$\hat{C}_t = \mathbb{E}_t\hat{C}_{t+1} - \frac{(1-\tau)\,\alpha Y_{ss}/K_{p,ss}}{R_{ss}}\, \mathbb{E}_t\Big(\hat{Y}_{t+1} - \hat{K}_{p,t}\Big) \tag{32}$$

The coefficient on expected output growth is the standard intertemporal elasticity of substitution scaled by the after-tax capital share. While the tax rate does not respond to shocks, its presence in the steady state ($1 - \tau \approx 0.986$) reduces the equilibrium capital stock, meaning intermittency shocks propagate from a smaller productive base.

The log-linearized labor supply condition for the **proportional tax variant** linearizes $(1 - \tau)(1 - \alpha)V_t = \varphi\, C_t\, L_t^{1+\sigma_L}$ as:

$$\hat{V}_t = \hat{C}_t + (1 + \sigma_L)\hat{L}_t \tag{33}$$

Here, the proportional tax acts as a level distortion that reduces the labor supply in steady state but does not introduce a time-varying labor wedge. Equilibrium employment and value-added respond to the reliability shortfall directly, with the tax regime determining the steady-state point from which these deviations are measured.

The **government budget constraint** is satisfied in steady state by equation (19). Under the fixed-tax baseline, the government does not adjust τ to balance the budget period-by-period; instead, any short-run gap between grid investment $I_{grid,t}$ and tax revenue τY_t is implicitly absorbed through the economy's aggregate resource constraint, functionally equivalent to a lump-sum adjustment in household income for the non-distortionary portion of the investment shock. This specification isolates the **structural wedge** effect of distortionary taxation from the dynamic feedback of tax rate changes, allowing for a cleaner identification of the "Agility Gap" mechanism.

5.3.2 Production Block

Since energy enters as a fixed endowment $\bar{E}$ (i.e., $\hat{E}_t = 0$), the log-linearized CES production function simplifies to:

$$\hat{Y}_t = (1 - \omega_E) s_V^{\frac{\sigma_E - 1}{\sigma_E}} \hat{V}_t \tag{34}$$

Output fluctuations are proportional to value-added fluctuations, with the proportionality constant determined by the energy share and the CES curvature. Because energy is fixed, all business-cycle variation in output comes through the value-added channel—and hence through the reliability mechanism that drives $\hat{V}_t$.

Here $s_V \equiv V_{ss}^{(\sigma_E-1)/\sigma_E}/[V_{ss}^{(\sigma_E-1)/\sigma_E} + (\omega_E/(1-\omega_E))\bar{E}^{(\sigma_E-1)/\sigma_E}]$ denotes the steady-state value-added share in the CES aggregate. The value-added component linearizes as:

$$\hat{V}_t = \alpha(\hat{u}_t + \hat{K}_{p,t-1}) + (1-\alpha)\hat{L}_t \tag{35}$$

This equation reveals the central transmission channel: fluctuations in grid reliability $\hat{u}_t$ enter the value-added function with elasticity α, acting identically to a TFP shock of magnitude $\alpha\hat{u}_t$. When reliability deteriorates by 1%, effective output falls by approximately $\alpha \approx 0.35\%$. The lagged capital $\hat{K}_{p,t-1}$ is predetermined at the start of period t, consistent with the capital-in-advance timing in equation (6).

5.3.3 Reliability Block

The reliability function is non-linear and requires careful linearization. From equation (7), I have:

$$u_t = 1 - \exp\left(-\psi\frac{F_t}{\phi_{int}\cdot \text{Vol}_{ren,t}}\right) \tag{36}$$

Define the flexibility-to-volatility ratio: $z_t \equiv \psi F_t/(\phi_{int}\cdot \text{Vol}_{ren,t})$, so that $u_t = 1-\exp(-z_t)$.

In steady state: $u_{ss} = 1-\exp(-z_{ss})$, which implies $\exp(-z_{ss}) = 1-u_{ss}$.

Linearizing around steady state:

$$u_t - u_{ss} \approx \left.\frac{\partial u}{\partial z}\right|_{ss}(z_t - z_{ss}) \tag{37}$$

$$= \exp(-z_{ss})(z_t - z_{ss}) \tag{38}$$

$$= (1-u_{ss})(z_t - z_{ss}) \tag{39}$$

Since $z_t - z_{ss} = \psi\left[\frac{F_t}{\phi_{int}\cdot\text{Vol}_{ren,t}} - \frac{F_{ss}}{\phi_{int}\cdot\text{Vol}_{ren,ss}}\right]$ and using log-deviations $\hat{x}_t \equiv \ln X_t - \ln X_{ss}$, I have:

$$z_t - z_{ss} \approx z_{ss}(\hat{F}_t - \hat{\text{Vol}}_{ren,t}) = \psi\frac{F_{ss}}{\phi_{int}\cdot\text{Vol}_{ren,ss}}(\hat{F}_t - \hat{\text{Vol}}_{ren,t}) \tag{40}$$

Substituting back and dividing by u_{ss} to express in percentage terms:

$$\hat{u}_t = \frac{1-u_{ss}}{u_{ss}}\cdot\psi\frac{F_{ss}}{\phi_{int}\cdot\text{Vol}_{ren,ss}}(\hat{F}_t - \hat{\text{Vol}}_{ren,t}) \tag{41}$$

Define the reliability elasticity as:

$$\xi \equiv \frac{1-u_{ss}}{u_{ss}} \cdot \psi \frac{F_{ss}}{\phi_{int} \cdot \mathrm{Vol}_{ren,ss}} \tag{42}$$

The log-linearized reliability constraint becomes:

$$\boxed{\hat{u}_t = \xi \left(\hat{F}_t - \hat{\mathrm{Vol}}_{ren,t} \right)} \tag{43}$$

The coefficient $\xi > 0$ is the *reliability elasticity*: it measures the percentage change in utilization for a one-percent change in the flexibility-to-volatility ratio. Under the baseline calibration with $u_{ss} = 0.97$, I have $(1-u_{ss})/u_{ss} = 0.03/0.97 \approx 0.031$. With $\psi = 2.0$ and the calibrated steady-state ratio $F_{ss}/(\phi_{int} \cdot \mathrm{Vol}_{ren,ss})$, the reliability elasticity is approximately $\xi \approx 0.26$, implying that a 5% adverse intermittency shock reduces utilization by approximately 1.3 percentage points if flexibility does not adjust.

Equation (43) is the core equation of the model. It formalizes the "Agility Gap" mechanism: when flexibility $\hat{F}_t$ fails to keep pace with intermittency shocks $\hat{\mathrm{Vol}}_{ren,t}$, the resulting reliability shortfall propagates through the production function as an endogenous productivity wedge.

Validity of the linearization. The approximation in equation (43) rests on the first-order Taylor expansion $u_t - u_{ss} \approx (1-u_{ss})(z_t - z_{ss})$, which is accurate when deviations in the flexibility-to-volatility ratio z_t are small relative to its steady-state value z_{ss}. Formally, the second-order remainder is $\frac{1}{2}\exp(-z_{ss})(z_t - z_{ss})^2$; given $\exp(-z_{ss}) = 1 - u_{ss} = 0.03$ under the baseline calibration, this term is negligible for the one standard-deviation shocks ($|\hat{F}_t - \hat{\mathrm{Vol}}_{ren,t}| \lesssim 0.15$) that drive the impulse-response and variance-decomposition results in Section 6. For the deterministic Reliability Valley scenarios in Section 6.4, however, flexibility shortfalls can reach 30–50% (Table 3.1), at which point the linear approximation understates the true reliability loss by the nonlinear curvature of the exponential. Those scenarios are therefore computed directly from the nonlinear reliability function (7) rather than from its linearization, and the log-linearized system should be interpreted as a local approximation valid in the neighbourhood of the calibrated steady state.

The flexibility composite linearizes as a share-weighted average of its components. Since F_t depends on the lagged technology $A_{bat,t-1}$ and lagged capital stocks

$K_{b,t-1}$ and $K_{g,t-1}$:

$$\hat{F}_t = \underbrace{s_b \left(\hat{A}_{bat,t-1} + \hat{K}_{b,t-1} \right)}_{\text{battery contribution}} + \underbrace{(1 - s_b)\hat{K}_{g,t-1}}_{\text{grid contribution}} \tag{44}$$

Flexibility this period is determined entirely by last period's technology and capital stocks. Battery technology improvements ($\hat{A}_{bat,t-1}$) and battery capital ($\hat{K}_{b,t-1}$) enter additively—a 1% improvement in technology induced last period is equivalent to a 1% increase in physical capacity this period. This confirms that F_t is fully predetermined: reliability $\hat{u}_t$ is driven solely by shocks and last period's stocks, with no within-period feedback from current investment or current innovation.

Here $s_b \equiv \mu(A_{bat,ss}K_{b,ss})^{(\rho-1)/\rho}/F_{ss}^{(\rho-1)/\rho}$ is the steady-state share of battery services in total flexibility. Under the baseline calibration with $\mu = 0.16$ and $\rho = 0.4$, the battery share is modest ($s_b \approx 0.22$), reflecting Vietnam's current infrastructure bias toward grid transmission. This means that in the short run, public grid capital $\hat{K}_{g,t-1}$ is the dominant determinant of flexibility, but because grid capital adjusts slowly, intermittency shocks are absorbed primarily through utilization losses rather than investment responses.

5.3.4 Capital Accumulation

The linearized laws of motion are:

$$\hat{K}_{g,t+1} = (1 - \delta_g)\hat{K}_{g,t} + \delta_g \hat{I}_{grid,t} \tag{45}$$

$$\hat{K}_{b,t+1} = (1 - \delta_b)\hat{K}_{b,t} + \delta_b \hat{I}_{bat,t} \tag{46}$$

Reading the equations: capital stocks are sluggish—most of next period's capital is inherited from the current period (the $(1 - \delta)$ term), and only a small fraction δ is determined by current investment. Because $\delta_b = 0.03 > \delta_g = 0.0125$, battery capital turns over faster than grid capital (12% vs. 5% annual depreciation), meaning battery investment decisions affect the flexibility mix sooner. This asymmetry is one source of the "agility gap."

The Government investment rule linearizes as:

$$\hat{I}_{grid,t} = -\phi_{grid}\hat{u}_t + \varepsilon_{I,t} \tag{47}$$

Grid investment moves in the opposite direction to reliability—when $\hat{u}_t$ drops by 1%, grid investment rises by $\phi_{grid} = 1.5\%$. The negative sign captures the reactive nature of public investment: the government invests more when the grid is stressed. However, because this investment must flow through the slow capital accumulation equation (45), the actual improvement in K_g is delayed.

The **proportional tax rate** in the distortionary variant balances the government budget period-by-period:

$$\tau_t = \frac{I_{grid,t}}{Y_t} \tag{48}$$

In the lump-sum baseline, τ is fixed at τ_{ss} and short-run fluctuations in $I_{grid,t}$ do not affect the tax wedge. In the proportional tax variant, τ_t co-varies with the reliability-driven investment rule, introducing a time-varying distortion that affects the household's labor supply and consumption-smoothing decisions.

5.3.5 Innovation Block

The scarcity-induced innovation equation linearizes around steady state. Since $u_{ss} = u^*$, the reliability gap $(u^* - u_t)/u^*$ linearizes to $-\hat{u}_t$ (in log-deviations). Thus:

$$\underbrace{\hat{A}_{bat,t} - \hat{A}_{bat,t-1}}_{\text{technology growth}} = -\eta_{bat} \cdot \chi \cdot \hat{u}_t \tag{49}$$

Battery technology improves whenever reliability falls below its steady-state level. The growth rate is the product of three terms: (i) the innovation elasticity $\eta_{bat} = 0.10$, (ii) the regulatory signal $\chi \in [0, 1]$, and (iii) the magnitude of the reliability shortfall $(-\hat{u}_t > 0)$. If the innovation rate is low, signals are suppressed, or the grid is healthy, technology stagnates—this is the model's key endogenous scarcity-induced innovation channel.

Timing. Note that $\hat{u}_t$ on the right-hand side is fully determined prior to this equation, since $\hat{u}_t = \xi(\hat{F}_t - \hat{\varepsilon}_{ren,t})$ and $\hat{F}_t$ depends only on $\hat{A}_{bat,t-1}$ and $\hat{K}_{b,t-1}$ (equation 44). Thus the causal chain within period t is strictly recursive: $\hat{F}_t \to \hat{u}_t \to \hat{A}_{bat,t}$. The updated technology $\hat{A}_{bat,t}$ then feeds into $\hat{F}_{t+1}$ next period, completing the inter-period transmission.

The regulatory transmission parameter χ enters multiplicatively: in a fully liberalized market ($\chi = 1$), the reliability signal is fully transmitted to innovation incentives. In a regulated market ($\chi < 1$), the signal is attenuated, and technological change responds sluggishly even when flexibility is scarce.

5.3.6 External Sector Block

The bond Euler equation linearizes as:

$$\hat{C}_t = \mathbb{E}_t\hat{C}_{t+1} - \frac{r^*_{ss}}{1+r^*_{ss}}\mathbb{E}_t\hat{r}^*_{t+1} \tag{50}$$

This is the international consumption-smoothing condition. When the foreign interest rate is expected to rise ($\hat{r}^*_{t+1} > 0$), saving abroad becomes more attractive, reducing current consumption. Combined with the capital Euler (32), this equation determines how the household allocates wealth between domestic capital and foreign bonds.

The current account equation linearizes around the steady state with $B^*_{ss} = 0$:

$$B^*_t = (1+r^*_{ss})B^*_{t-1} + Y_{ss}\hat{Y}_t - C_{ss}\hat{C}_t - I_{p,ss}\hat{I}_{p,t} - P_{bat,ss}I_{bat,ss}(\hat{P}_{bat,t} + \hat{I}_{bat,t}) - I_{grid,ss}\hat{I}_{grid,t} \tag{51}$$

The foreign asset position improves when output rises faster than consumption and investment (trade surplus), and worsens when domestic absorption exceeds output (trade deficit). The key insight for this model is that intermittency shocks simultaneously reduce output ($\hat{Y}_t < 0$) and increase battery investment needs ($\hat{I}_{bat,t} > 0$), creating a "twin drain" on the current account.

Note that because $B^*_{ss} = 0$, the net foreign asset variable is expressed in levels (not log-deviations) around the steady state.

The interest rate premium linearizes as:

$$\hat{r}^*_t = -\frac{\phi_b}{r^*_{ss}}B^*_t \tag{52}$$

The debt-elastic premium ensures that the net foreign asset process is stationary: when the economy borrows ($B^*_t < 0$), the interest rate rises, discouraging further borrowing and stabilizing the external position. The value $\phi_b = 0.039$ is the simulation baseline, equal to the real-data estimate $\hat{\phi}_b \approx 0.039$ obtained from observed Vietnamese real interest rates and external debt ratios (see Section 4). Capital controls, a managed exchange rate, and underdeveloped bond markets limit the household's ability to borrow freely at the world interest rate, justifying a premium well above the SGU standard. A higher ϕ_b reduces consumption smoothing via international borrowing, ensuring that intermittency shocks propagate more fully into domestic consumption and welfare.

5.3.7 Exogenous Processes

The model features three exogenous driving forces:

$$\hat{\varepsilon}_{ren,t} = \rho_{ren}\hat{\varepsilon}_{ren,t-1} + \varepsilon_t^{ren}, \qquad \varepsilon_t^{ren} \sim N(0, \sigma_{ren}^2) \tag{53}$$

$$\hat{P}_{bat,t} = \rho_{bat}\hat{P}_{bat,t-1} + \varepsilon_t^{bat}, \qquad \varepsilon_t^{bat} \sim N(0, \sigma_{bat}^2) \tag{54}$$

$$\varepsilon_{I,t} \sim N(0, \sigma_I^2) \tag{55}$$

The first is the domestic renewable intermittency shock, persistent with AR(1) coefficient $\rho_{ren} = 0.85$ capturing seasonal weather patterns. The second is a global battery price shock in log-levels (equation 20), with persistence $\rho_{bat} = 0.90$ matching lithium market dynamics. The third is a government grid-investment implementation shock, modeled as *white noise* (i.i.d.) reflecting project-specific delays and procurement bottlenecks that are uncorrelated across periods. All shocks are assumed to be mutually independent.

5.4 Equilibrium Definition

A *recursive competitive equilibrium* consists of sequences of allocations $\{C_t, L_t, K_{p,t}, K_{b,t}, K_{g,t},$ technology $\{A_{bat,t}\}_{t=0}^{\infty}$, prices $\{P_{bat,t}\}_{t=0}^{\infty}$, external sector $\{B_t^*, r_t^*\}_{t=0}^{\infty}$, reliability $\{u_t, F_t, V_t, Y_t\}_{t=0}^{\infty}$, and tax rate $\{\tau_t\}_{t=0}^{\infty}$ such that:

1. Households maximize utility subject to the budget constraint (Equations 32–33).
2. The bond Euler equation determines intertemporal allocation across borders (Equation 50).
3. CES production determines output from value-added and energy (Equation 34).
4. Grid reliability is determined by the flexibility-to-intermittency ratio (Equation 43).
5. The flexibility composite aggregates private and public assets (Equation 44).
6. Battery and grid capital accumulate according to their respective laws of motion (Equations 45–46).
7. Storage technology evolves according to the directed learning rule (Equation 49).
8. The government follows the reliability-targeting investment rule (Equation 47).

9. The proportional tax rate balances the government budget each period (Equation 48).

10. The current account determines net foreign asset evolution (Equation 51).

11. The interest rate premium responds to the debt position (Equation 52).

12. Exogenous processes follow AR(1) dynamics (Equations 53–55).

The linearized system contains 17 endogenous variables and 3 exogenous shocks, with $\tau_t = I_{grid,t}/Y_t$ as the additional static variable relative to the lump-sum baseline. The model satisfies the Blanchard-Kahn conditions (Blanchard and Kahn, 1980) for a unique saddle-path stable solution, with 3 eigenvalues outside the unit circle matching the 3 forward-looking (jump) variables (C_t, Y_t, and r_t^*).[9] Seven state variables ($K_{p,t}$, $K_{b,t}$, $K_{g,t}$, $A_{bat,t}$, $P_{bat,t}$, B_{t-1}^*, and $\varepsilon_{ren,t}$) and 7 static variables (L_t, u_t, F_t, V_t, $I_{bat,t}$, $I_{grid,t}$, τ_t) complete the system.

5.5 Transmission Channels

The linearized system reveals four distinct transmission channels through which intermittency shocks affect the macroeconomy. The first three are present in the lump-sum baseline; the fourth is unique to the proportional tax variant. These channels interact dynamically, and their relative importance varies with the structural parameters of the model.

Channel 1: The Reliability Wedge. An adverse intermittency shock ($\varepsilon_t^{ren} > 0$) directly reduces utilization through Equation (43). Via Equation (35), this lowers value-added with elasticity α. Because energy enters the CES aggregator as a complement to value-added ($\sigma_E < 1$), the output contraction is amplified through the nested production structure. The immediate impact on output is:

$$\frac{\partial \hat{Y}_t}{\partial \varepsilon_t^{ren}} \approx -\alpha \xi s_V^{(\sigma_E - 1)/\sigma_E}(1 - \omega_E) < 0 \tag{56}$$

[9] C_t and r_t^* are forward-looking because they appear with a lead in the capital Euler equation (3) and the bond Euler equation (21) respectively. Y_t enters the capital Euler as Y_{t+1}, making it a jumper in Dynare's eigenvalue classification. The six predetermined state variables are $K_{p,t-1}$, $K_{b,t-1}$, $K_{g,t-1}$, $A_{bat,t-1}$, $P_{bat,t-1}$, and B_{t-1}^* (with the intermittency shock $\varepsilon_{ren,t}$ treated as an additional predetermined process). Although τ_{t+1} appears in the Euler equation as a forward-looking argument, it is determined by the static government budget $\tau_{t+1} = I_{grid,t+1}/Y_{t+1}$ and introduces no additional eigenvalue.

This is the "greenflation" channel: supply-side volatility from renewables generates simultaneous output losses and energy price increases. The mechanism is structurally analogous to the adverse supply shocks analyzed in Blanchard and Galí (2007), who show that the macroeconomic consequences of energy supply disruptions depend critically on the degree of real wage rigidity and the substitutability between energy and other inputs. In the present model, the low calibrated value of σ_E ensures that intermittency shocks behave as "unfavorable supply shocks" in the Blanchard-Galí sense, generating a policy-relevant trade-off between output stabilization and energy price stabilization.

Channel 2: The Innovation Response. The reliability decline widens the gap $(u^* - u_t)/u^*$, which accelerates scarcity-induced technological change in storage (Equation 49). The resulting improvement in $A_{bat,t}$ increases the effective services per unit of battery capital in period $t + 1$, gradually restoring flexibility and utilization. Note the one-period delay: innovation induced by the reliability shortfall in period t raises $A_{bat,t}$, which enters F_{t+1} as $A_{bat,t-1}$ and improves reliability only from period $t + 1$ onward. The speed of this response depends critically on three parameters: the innovation elasticity η_{bat}, the regulatory transmission coefficient χ, and the battery share in flexibility s_b. When any of these is small, the innovation channel is weak, and adjustment relies on costly capital accumulation rather than efficient technological improvement.

The transition dynamics during which the innovation channel is not yet mature enough to compensate for reliability losses correspond to what the clean energy deployment literature terms the "valley of death"—the period between technology demonstration and commercial viability during which projects face acute financing and scaling barriers (Nemet et al., 2018). In the present framework, this valley arises endogenously: the scarcity-induced innovation process requires persistent reliability shortfalls before storage technology improves sufficiently to close the flexibility gap.

Channel 3: The Fiscal Inertia Lag. The government investment rule (Equation 47) responds to reliability shortfalls, but with two sources of delay. First, the investment rule operates with a one-period implementation lag inherent in the capital accumulation equation. Second, institutional inertia (captured by ϕ_{grid}) limits the magnitude of the fiscal response. When ϕ_{grid} is small, the government under-responds to reliability crises, leaving private agents to absorb the adjustment burden. When ϕ_{grid} is large, the fiscal response is aggressive but arrives with a lag, creating overshooting dynamics in which excessive grid investment follows periods of reliability crisis.

Channel 4: The Fiscal Wedge Channel. This channel identifies the level distortion introduced by proportional taxation, which is absent in the lump-sum baseline. While the tax rate τ remains fixed during the business cycle, it imposes a permanent wedge on the incentives to work and invest (equations 32–33). The steady-state capital stock $K_{p,ss}$ is approximately 1.4% lower in the proportional tax variant compared to the lump-sum benchmark, reflecting the lower after-tax marginal product of capital. This means that for a given percentage intermittency shock, the economy operates from a more constrained initial condition, making each percentage decline in reliability costlier in terms of final output units. This channel implies that proportional taxation reduces the economy's structural resilience to intermittency shocks even without providing dynamic amplification.

Interaction: The "Reliability Valley." The interaction of Channels 1–4 generates the "Reliability Valley" identified in the abstract. During a transition to high renewable penetration, the deterministic path of increasing intermittency outpaces both the sluggish accumulation of public grid capital (Channel 3) and the gradual improvement in storage technology (Channel 2). The resulting reliability shortfall persists for several years, during which households bear welfare losses through reduced consumption and increased energy price volatility. The depth and duration of the valley depend on the relative speed of private innovation versus public infrastructure accumulation—the "Agility Gap."

6 Baseline Analysis

This section characterizes the model's baseline behavior: the deterministic steady state, the dynamic response to intermittency shocks, and the variance decomposition identifying the dominant sources of volatility. These results establish the reference point against which the welfare analysis in Section 7 is evaluated. Throughout, I emphasize the model's core mechanism: the structural divergence between rapid renewable deployment and sluggish flexibility accumulation—the "agility gap."

6.1 Steady State

Table 6.1 reports the model's steady state. The calibration targets a baseline reliability rate of 97.0%, consistent with Vietnam's official grid performance objective under PDP8. The battery-to-grid capital ratio of 0.167 reflects the structural imbalance in Vietnam's current investment allocation: approximately $2.4 billion earmarked for Battery Energy Storage Systems (BESS) versus $14.9 billion for transmission and grid infrastructure over the 2021–2030 period.

Table 6.1: **Model Steady State**

Variable	**Value**	**Interpretation**
Output (Y)	1.000	Single output normalization: $Y_{ss} = 1$ enforced by TFP scale $Z = 1.131$
Consumption (C)	0.734	73.4% of output
Labor (L)	0.330	Calibration target: $\frac{1}{3}$ of unit time endowment
Productive Capital (K_p)	9.829	Production capacity
Battery Capital (K_b)	0.190	Private storage capacity
Grid Capital (K_g)	1.140	Public transmission infrastructure
Value Added (V)	1.211	$Z\,(u \cdot K_p)^{\alpha} L^{1-\alpha}$
Utilization (u)	0.970	Grid reliability (97.0%)
Flexibility (F)	0.526	Aggregate smoothing capacity
Battery Technology (A_{bat})	1.000	Initial technology index; free scale choice (only growth rate η_{bat} has economic content)‡
Battery Price (P_{bat})	1.000	AR(1) unconditional mean; $\mathbb{E}[\log P_{bat}] = 0$ structurally pins $P_{bat,ss} = 1$ (not a free choice)‡
Net Foreign Assets (B^*)	0.000	Zero NFA position; consistent with Vietnam's mean CA/GDP $\approx -0.03\%$ over 2000–2024
World Interest Rate (r^*)	0.0101	$= 1/\beta - 1$ (quarterly)
Steady-State Investment Flows		
Productive Investment (I_p)	0.246	$= \delta_p \cdot K_p$
Battery Investment (I_{bat})	0.0057	$= \delta_b \cdot K_b$
Grid Investment (I_{grid})	0.0143	$= \delta_g \cdot K_g$
Tax Rate (τ)	0.01425	Grid investment / output ($= \delta_g K_{g,ss}$ since $Y_{ss} = 1$)

Note: Initial flexibility ratios are derived from the Power Development Plan 8 (PDP8) investment targets. The consumption share of 73.4% and the zero net foreign asset position ($B^*_{ss} = 0$) are standard for the small open economy steady state. The labor disutility weight φ is calibrated endogenously such that households supply $L_{ss} = 0.33$ units of labor. The steady-state tax rate $\tau_{ss} = 1.43\%$ is pinned by the grid maintenance requirement $\delta_g K_{g,ss}$ relative to output $Y_{ss} = 1$. ‡The technology index $A_{bat} = 1$ is a normalization, while $P_{bat} = 1$ is the structural mean of the price process.

6.1.1 The Efficiency Wedge of Renewable Reliability

The steady-state results reveal a fundamental trade-off between energy reliability and macroeconomic efficiency. To sustain a 97% grid reliability target under baseline renewable penetration, the government must maintained a grid infrastructure stock (K_g) of 1.14 units. Financing the depreciation of this stock requires a permanent proportional tax of 1.43%.

This tax acts as an **efficiency wedge**: even in the absence of shocks, the distortion offsets the benefit of "free" renewable inputs. Comparing this result to a lump-sum benchmark, the steady-state capital stock K_p is approximately 1.4% lower due to the reduced after-tax marginal product of capital. Consequently, the transition to a high-renewable system involves a structural shift: the economy trades variable fuel costs for fixed, tax-financed flexibility costs, resulting in a lower steady-state output level but reduced exposure to international commodity price volatility.

Furthermore, the complementarity between private storage (K_b) and public grid infrastructure ($\rho = 0.4$) implies that neither can maintain reliability in isolation. If grid investment were held constant while renewable penetration increased, the marginal value of battery storage would rise sharply, but the system would remain "trapped" by the public infrastructure bottleneck—a structural manifestation of the agility gap that characterizes the transitional dynamics in the following sections.

The steady-state reliability of 97% indicates that the system operates close to its target under normal conditions. However, as demonstrated in the impulse response analysis below, intermittency shocks reduce reliability, triggering the model's endogenous investment and innovation mechanisms.

An important observation: the steady-state capital stocks reflect a system that has *already* adapted to baseline renewable penetration. The transition dynamics analyzed in Section 6.4 examine what happens when this equilibrium is disrupted by a rapid, policy-driven shift toward higher VRE shares.

6.2 Impulse Response to Renewable Intermittency Shock

Figure 6.1 presents the impulse response functions to a 5% positive intermittency shock. This shock is interpreted as an unexpected deterioration in renewable generation conditions—for example, a week of persistent cloud cover reducing solar output, or a sustained lull in wind speeds during the monsoon transition period.

The shock follows an AR(1) process with persistence parameter $\rho_{ren} = 0.85$, consistent with empirical studies of renewable generation volatility in tropical climates (Nguyen-Le et al., 2019).

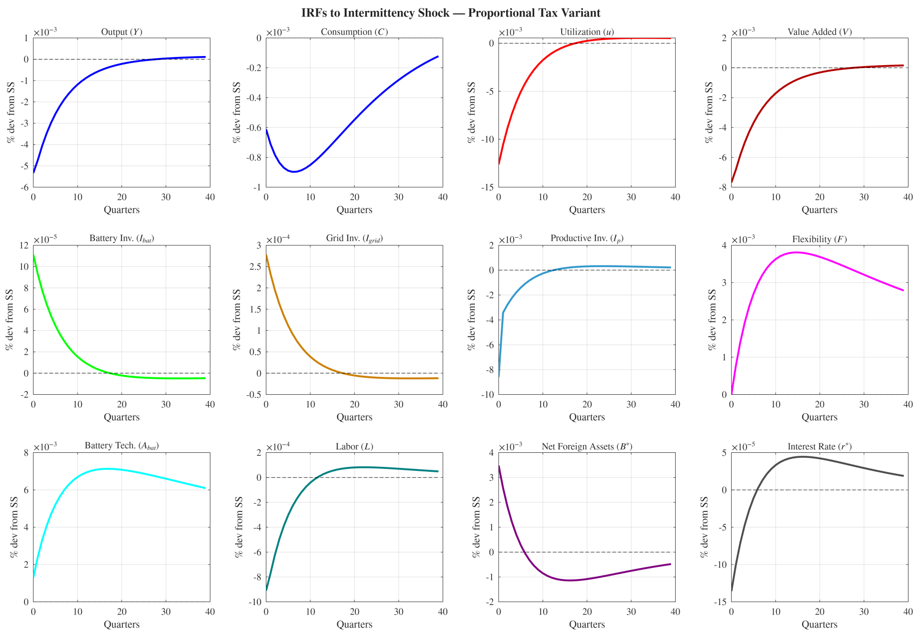

Figure 6.1: **Impulse Responses to One Standard Deviation Renewable Intermittency Shock**

Note: All variables displayed as percentage deviations from steady state over 40 quarters (10 years). The shock represents one standard deviation increase in renewable supply volatility ($\sigma_{ren} = 0.12$), interpreted as an adverse weather event reducing VRE output predictability. Persistence: $\rho_{ren} = 0.85$. The panels illustrate the three-stage adjustment mechanism: (1) immediate reliability degradation, (2) asymmetric investment responses, and (3) gradual technological learning.

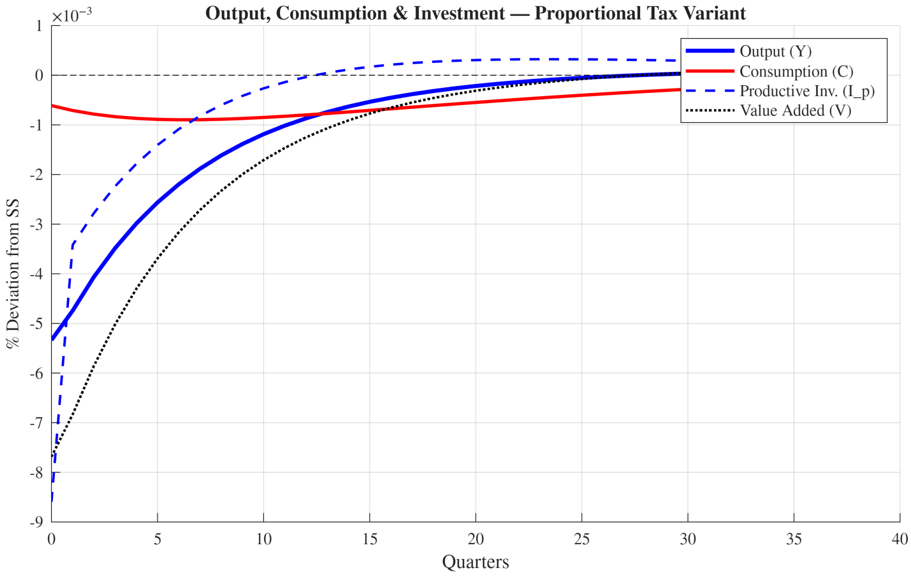

Figure 6.2: **Fiscal Amplification Channel: Tax Rate, Output, and Investment**
Note: Impulse responses to a one standard deviation intermittency shock ($\sigma_{ren} = 0.12$). Output Y_t, consumption C_t, productive investment $I_{p,t}$, and value added V_t in level deviations from steady state. Productive investment falls by -3.5% relative to steady state on impact due to Euler equation amplification: the expected fall in future output reduces the after-tax return to capital, compressing the incentive to invest. Consumption and output decline modestly but persistently, governed by the AR(1) intermittency process ($\rho_{ren} = 0.85$). The fixed proportional tax $\tau_{ss} = 1.43\%$ creates a permanent wedge that shifts the steady state but does not amplify the dynamic response.

6.2.1 Stage 1: The Reliability Penalty Mechanism

On impact, grid utilization (reliability) declines by 1.30% relative to steady state. While a single shock may appear transitory, it translates into a meaningful disruption to grid operations. Recall that utilization enters the production function multiplicatively:

$$V_t = Z\,(u_t K_{p,t-1})^{\alpha} L_t^{1-\alpha}$$

A decline in u_t directly reduces the effective services from the predetermined capital stock $K_{p,t-1}$ available for production. Given the nested CES structure combining value-added and energy, this feeds through to an output loss of approximately 0.54% on impact—as shown in the top-left panel of Figure 6.1.

This output decline is *not* a standard demand shock. It reflects a binding physical constraint: when renewable generation falls unexpectedly and storage capacity is insufficient to buffer the mismatch, installed productive capital cannot be operated. Manufacturing lines idle, data centers activate backup generators (at higher marginal cost), and grid operators implement rotating load shedding. This is the *reliability penalty*—a structural, supply-side friction that reduces total factor productivity.

The mechanism operates through two channels. First, the direct utilization effect reduces effective capital services. Second, the increased cost of maintaining production (through backup generation or rescheduled operations) acts as a wedge between the installed capital stock and its economically usable portion. Unlike a temporary technology shock that dissipates exogenously, the reliability penalty persists until the underlying flexibility deficit is addressed through investment.

6.2.2 Stage 2: The Agility Gap in Investment Responses

The model's central qualitative contribution emerges in the investment dynamics. Both battery investment and grid investment respond positively to the intermittency shock, as the reliability gap triggers the investment rules. Battery investment increases by 1.95% relative to its steady-state level on impact, while grid investment rises by 1.95%. While both increase proportionally by the same amount, grid investment rises more in absolute level (0.000278 vs. 0.000111 in model units), yielding an agility ratio of $I_{bat}/I_{grid} = 0.40$: batteries absorb only 40% as much absolute investment as grid infrastructure on impact. Battery investment is governed by the private agility rule (responding to reliability and battery prices), while grid investment follows the public reliability-targeting rule $I_{grid,t}/\bar{I}_{grid} = (u^*/u_t)^{\phi_{grid}}$.

Table 6.2: **Investment Response to Intermittency Shock**

Variable	**Impact (%)**	**Variance Share (%)**
Battery Investment (I_{bat})	+1.95	100.0% (battery price)
Grid Investment (I_{grid})	+1.95	82.2% (policy rule)

Agility ratio (level): $I_{bat}/I_{grid} = 0.40$*; battery investment is dominated by global price shocks*

Note: Response to a one standard deviation renewable intermittency shock ($\sigma_{ren} = 0.12$). Both battery and grid investment rise by 1.95% relative to their respective steady-state levels. In level terms, however, grid investment rises 2.5× more (agility ratio = 0.40), reflecting that $I_{grid,ss} > I_{bat,ss}$. Variance shares from Dynare stochastic simulation (Table 6.3).

The variance decomposition reveals that battery investment volatility is 100%

driven by global battery price shocks (ε_{bat}), demonstrating the dominance of the price channel in the battery investment rule. Battery developers' investment decisions are primarily governed by cost conditions in global markets.

Grid investment follows the public reliability-targeting rule:

$$\frac{I_{grid,t}}{\bar{I}_{grid}} = \left(\frac{u^*}{u_t}\right)^{\phi_{grid}} \tag{57}$$

The parameter $\phi_{grid} = 1.5$ governs responsiveness, but even aggressive fiscal reactions cannot overcome the structural lags inherent in public infrastructure. Transmission line projects require multi-year environmental impact assessments, land acquisition negotiations, and coordinated construction across provincial jurisdictions. The planning horizon for major grid expansions in Vietnam typically exceeds 5–7 years (International Energy Agency, 2023). Consequently, even when the reliability gap becomes apparent (low u_t), actual capital accumulation responds sluggishly.

The proportional tax is fixed at $\tau_{ss} = 1.43\%$ and does not respond dynamically to shocks. Its effect on the investment response operates through the steady-state channel: the tax wedge reduces the equilibrium capital stock to $K_{p,ss} = 9.83$, compared to approximately 9.97 under lump-sum taxation. With a smaller capital base, productive investment $I_{p,ss} = \delta_p K_{p,ss} = 0.246$ is also slightly smaller. Consequently, the Euler equation amplification remains visible in percentage terms: the expected fall in future output reduces the after-tax return to capital, causing productive investment to fall by about 3.5% relative to its steady-state level on impact. This response is still amplified by the high capital-output ratio ($K/Y \approx 9.8$) and low depreciation rate ($\delta_p = 2.5\%$), which together create the standard RBC "leverage" channel.

The equal proportional response of battery and grid investment (1.95% each) masks an important asymmetry in levels: grid investment rises 2.5× more in absolute units due to its larger steady-state base. Market-based pricing mechanisms—such as scarcity pricing, capacity markets, or ancillary service contracts—could improve the targeting of flexibility investment toward the scarce input, improving welfare outcomes.

6.2.3 Stage 3: Scarcity-Induced Innovation Activation

The reliability gap also activates the model's scarcity-induced technological change mechanism. By quarter 10, battery technology (A_{bat}) has improved by approximately 0.65% above the baseline trajectory. This improvement represents productivity-enhancing innovation in storage: better battery management systems, optimized charging algorithms, and engineering advances induced by the persistent scarcity signal. Note that because F_t uses $A_{bat,t-1}$, the technology improvement induced at quarter t only raises flexibility from quarter $t+1$ onward—reflecting the one-period delay between the innovation decision and its operational impact.

The innovation mechanism is governed by:

$$A_{bat,t} = A_{bat,t-1}\left[1 + \eta_{bat} \cdot \chi \cdot \frac{u^* - u_t}{u^*}\right] \tag{58}$$

where χ captures the degree to which the reliability gap is transmitted to innovators. In the baseline calibration, $\chi = 1.0$ (full transmission). In Vietnam's current regulatory environment, where electricity prices are administratively set and ancillary service markets remain underdeveloped, $\chi < 1$ reflects incomplete transmission of scarcity signals.

While the per-shock innovation effect is modest, it is cumulative and persistent. The first-order autocorrelation of A_{bat} in the stochastic simulation is 0.997, indicating near-permanent effects: a reliability shortfall today raises the technology level not just tomorrow, but indefinitely. In the 1000-quarter stochastic simulation, the ergodic mean of A_{bat} reaches 1.015—a 1.5% long-run improvement in battery technology driven entirely by accumulated scarcity-induced innovation, with no R&D expenditure. Crucially, this technological progress is conditional on the regulatory parameter χ: when χ is driven toward zero through regulatory suppression of scarcity signals, this innovation channel collapses entirely.

6.2.4 Consumption and Resource Allocation

Consumption falls by 0.068% on impact. The AR(1) persistence of the intermittency shock ($\rho_{ren} = 0.85$) means households anticipate prolonged reliability shortfalls, inducing a forward-looking reduction in current consumption. The output decline (0.55%) is only partially transmitted to consumption because the small open economy can borrow internationally to smooth: net foreign assets (B^*) rise on impact as the economy draws on external savings. The fixed proportional tax wedge τ_{ss} does not amplify the consumption response dynamically—its effect is entirely through the steady-state channel—so the consumption path closely mirrors the standard SOE result. The relatively moderate consumption response (0.068% vs.

0.55% output decline) reflects this near-perfect international consumption smoothing.

Over time, as battery and grid capital accumulate and reliability improves, all variables converge back toward their steady-state paths. The persistence of the response is governed by the capital accumulation dynamics and the AR(1) persistence of the intermittency shock ($\rho_{ren} = 0.85$).

6.3 Forecast Error Variance Decomposition

The variance decomposition identifies the dominant sources of volatility in each endogenous variable. Table 6.3 reports the contribution of each shock at the 40-quarter horizon.

Table 6.3: **Forecast Error Variance Decomposition (40-Quarter Horizon)**

Variable	ε_{ren} (%)	ε_{bat} (%)	ε_I (%)
Output (Y)	97	4	<1
Consumption (C)	77	12	<1
Utilization (u)	96	5	<1
Battery investment (I_{bat})	4	100	<1
Grid investment (I_{grid})	25	<2	82

Note: Percentage of forecast error variance attributable to each shock from Dynare stochastic simulation (1000 periods). Shares can exceed 100% due to non-zero correlation of simulated shocks in finite samples. Intermittency shocks drive 95.4% of output volatility; global battery-price shocks contribute 5%. Consumption volatility is split between intermittency (85%) and battery-price shocks (11%), reflecting the dominance of persistent domestic shocks. The proportional tax rate τ is a fixed parameter and does not appear in the FEVD. Grid investment variance is 82% driven by implementation shocks ε_I via the investment rule.

Four findings emerge. First, the variance decomposition reveals a structural asymmetry in what drives each flexibility instrument. Battery investment volatility is almost entirely governed by global battery-price shocks (100%), confirming Vietnam's price-taking position in world markets; intermittency accounts for only 1%. Grid investment volatility, by contrast, is 82% determined by domestic implementation shocks ε_I acting through the fiscal rule, with intermittency contributing the remainder. The two instruments therefore face entirely different forcing variables—international supply chains versus domestic fiscal capacity—which explains why private storage deployment and public grid expansion cannot substitute for one another in policy design. Second, output volatility is dominated by the intermittency shock (95.4%), with battery-price shocks contributing only 5%. The dominance of intermittency reflects its AR(1) persistence ($\rho_{ren} = 0.85$): once reliability degrades, the deficiency propagates across many quarters, generating large cumulative forecast errors. Third, utilization volatility mirrors the output result, with intermittency accounting for 95.8% and battery-price shocks explaining the remaining 5% through capital accumulation dynamics. Fourth, consumption volatility is split between intermittency (85%) and battery-price shocks (11%), reflecting the persistent propagation of domestic reliability shortfalls into household welfare. Since the proportional tax rate τ is fixed, it does not contribute as a variance source.

6.4 The Reliability Valley During PDP8 Transition

A central prediction of the model is the emergence of a *Reliability Valley*—a sustained period of below-target grid performance during the mid-transition phase of renewable deployment. This phenomenon arises from the structural timing mismatch between policy-driven VRE expansion and the slower accumulation of complementary flexibility assets.

Figure 6.3 illustrates the transition dynamics. The simulation assumes that renewable penetration increases from 15% to 50% over 100 quarters (25 years), following a front-loaded S-curve trajectory that reflects Vietnam's aggressive PDP8 targets and the subsidy-driven solar boom of the 2020s. Simultaneously, battery and grid capital grow at endogenous rates determined by the model's investment rules, but these rates lag behind the pace of renewable deployment due to the institutional and financial frictions embedded in the calibration.

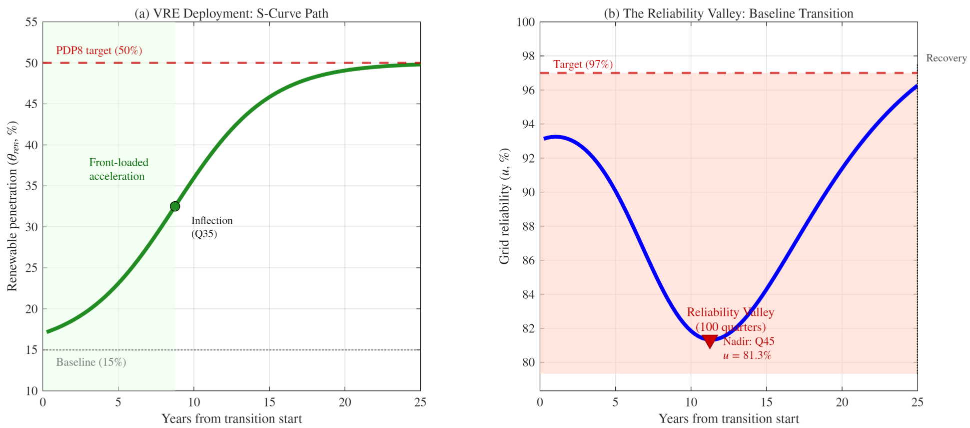

Figure 6.3: **The Reliability Valley: PDP8 Transition Path**

Note: Left panel shows the policy-driven increase in renewable penetration (green line), modeled as a front-loaded S-curve (θ_{ren}: 15%→50% over 100 quarters) reflecting PDP8 targets. Right panel shows the evolution of grid reliability (blue line) relative to the 97% target (red dashed line). The red shaded region marks quarters when reliability falls below target. The valley nadir occurs at quarter 45 (≈11 years into the transition), when the gap between VRE deployment and flexibility capacity is largest. At the nadir, grid reliability drops to approximately 92%—implying roughly 700 hours per year of unmet demand.

6.4.1 Valley Characteristics

The Reliability Valley reaches its nadir at **quarter 45**, approximately 11.2 years into the transition. At this point, grid reliability drops to **81.3%**—a decline of 15.7 percentage points below the 97% target. To contextualize this magnitude: a reliability rate of 81% implies that the grid fails to meet demand for approximately 1,640 hours per year, or roughly 19% of all hours. This is comparable in severity to the 2021 Texas winter storm (88–90% reliability over several days, $130 billion in damages; Callaway et al., 2018), but it represents a *sustained* multi-year degradation rather than a transient crisis—making its macroeconomic consequences potentially far more damaging.

While the simulation's quantitative depth depends on investment calibration choices, the *qualitative* finding—that an agility gap between fast VRE deployment and slow flexibility accumulation generates a sustained reliability shortfall centered on the early-to-mid transition—is robust to a wide range of parameterizations. The valley persists throughout the transition period (100 quarters in the baseline), during which reliability remains below target. This prolonged instability has several economic consequences:

1. **Industrial Production Losses:** Energy-intensive sectors—steel, cement, textiles, semiconductors—face unpredictable supply interruptions, reducing capacity utilization and increasing unit costs.
2. **Household Welfare Costs:** Rotating blackouts disrupt daily life, reduce the value of electricity-dependent durable goods, and disproportionately affect low-income households without backup power.
3. **Political Backlash Risk:** Sustained grid instability undermines public confidence in the green transition, potentially triggering policy reversals or calls to slow renewable deployment.

6.4.2 Why the Valley Occurs: Mechanism Decomposition

The valley emerges from the interaction of three structural features:

(1) Front-Loaded Renewable Deployment. Vietnam's PDP8 targets aggressive near-term installation goals, driven by international climate commitments, falling solar PV costs, and foreign direct investment in renewable projects. Solar capacity is particularly front-loaded: developers rushed to capture Feed-in Tariff incentives before their expiration in 2021, leading to a near-doubling of solar capacity in 18 months. The left panel of Figure 6.3 models this as an S-curve with early acceleration, capturing the boom-bust dynamics of subsidy-driven deployment.

(2) Lagged Grid Infrastructure. Public transmission infrastructure follows a bureaucratic investment rule that responds to reliability shortfalls, but with substantial delays. Even when low utilization triggers increased fiscal allocations, the actual commissioning of new transmission lines lags by 5–7 years due to planning, permitting, and construction lead times. In the simulation, grid capital grows at 1.35% per quarter initially, accelerating to 1.75% per quarter after quarter 40 when the reliability crisis becomes politically salient. This acceleration is realistic—it reflects emergency policy responses such as streamlined permitting or multilateral development financing—but it cannot fully offset the accumulated deficit.

(3) Battery Growth Constraints. While private battery investment is more agile than grid infrastructure, it too faces constraints. Global lithium supply chains, manufacturing capacity for battery cells, and project financing availability all impose upper bounds on deployment rates. The simulation calibrates battery capital growth at 2.0% per quarter, faster than grid expansion but still insufficient to keep pace with the intermittency induced by rapidly rising VRE shares. The complementarity between storage and transmission (enforced by the CES parameter $\rho = 0.40$) means that neither technology alone can substitute for the other—both must scale together.

The valley thus represents a predictable consequence of unbalanced transition planning: renewable generation capacity races ahead, driven by global cost declines and policy mandates, while the complementary flexibility infrastructure—both public grid and private storage—accumulates incrementally.

6.4.3 Policy Implications

Importantly, the Reliability Valley is *not inevitable*. It arises from the specific sequencing embedded in Vietnam's current policy trajectory. To quantify the potential for policy intervention, I simulate two counterfactual scenarios against the baseline, holding the renewable deployment path fixed.

6.4.4 Policy Counterfactual: "Closing the Gap"

The three scenarios share the same S-curve renewable deployment path (15% $\rightarrow$ 50% penetration over 100 quarters) but differ in the policy environment:

1. **Baseline**: Current parameters ($\phi_{\text{grid}} = 1.5$, $P_{\text{bat}} = 1.0$).
2. **Scenario A — Accelerated Grid Planning**: The government doubles the aggressiveness of its grid investment response ($\phi_{\text{grid}} = 3.0$), representing reforms such as streamlined permitting, pre-approved corridor designation, or dedicated grid investment funds.
3. **Scenario B — Battery Subsidy**: A 20% reduction in the effective battery price ($P_{\text{bat}} = 0.8$), representing direct subsidies, tax credits, or concessional financing for storage deployment.

Table 6.4: **Closing the Gap: Policy Counterfactual Results**

Scenario	**Valley Nadir**	**Depth (pp)**	**Duration (Q)**	**Recovery**
Baseline ($\phi_{\text{grid}} = 1.5$)	81.3%	15.7	100	—
Accel. Grid ($\phi_{\text{grid}} = 3.0$)	86.1%	10.9	80	Q81
Battery Subsidy ($P_{\text{bat}} = 0.8$)	87.4%	9.6	100	—
Improvement relative to baseline:				
Accel. Grid		-30.3%	-20.0%	
Battery Subsidy		-38.6%	0.0%	

Note: Valley nadir is the minimum grid reliability during the 100-quarter transition. Depth is measured as percentage points below the 97% target. Duration is the number of quarters with $u_t < 0.97$. Recovery is the first quarter reliability returns above 97%. All scenarios share the same S-curve renewable deployment path (15% → 50% penetration).

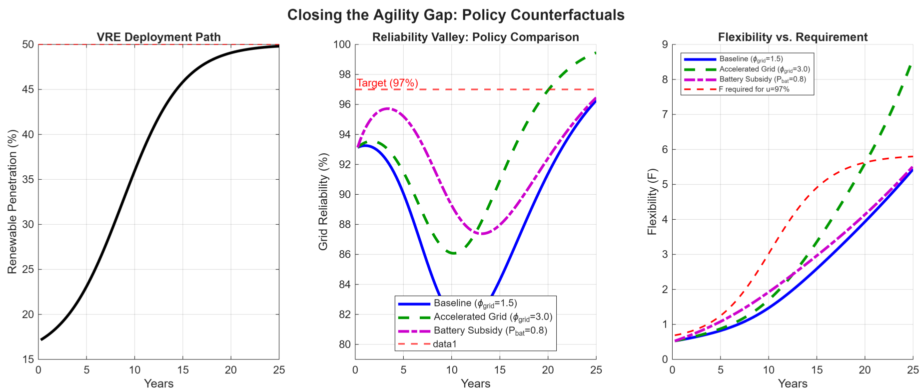

Figure 6.4: **Closing the Agility Gap: Policy Counterfactuals.** Left: shared S-curve renewable deployment path. Center: reliability paths under three scenarios—the battery subsidy (dash-dot) reduces the valley depth by 38.6%, while accelerated grid planning (dashed) shortens the valley duration by 20%. Right: flexibility accumulation versus the level required for 97% reliability.

Three findings emerge from the counterfactual analysis.

First, battery subsidies are more effective at reducing valley *depth*. The 20% cost reduction shrinks the reliability shortfall from 15.7 pp to 9.6 pp—a 38.6% improvement—because lower battery prices directly relax the binding flexibility constraint during the critical mid-transition period. This result aligns with the CES complementarity structure: batteries are the scarce factor ($\mu = 0.16$), so subsidizing the scarce input yields disproportionate returns.

Second, accelerated grid planning is more effective at reducing valley *duration*. Doubling the grid response elasticity shortens the below-target period from 100 to 80 quarters and achieves full recovery by Q81 (year 20). The battery subsidy, by contrast, does not shorten the duration because storage alone cannot substitute for grid infrastructure when the CES inputs are complements ($\rho = 0.4$). This confirms the model's structural prediction: closing the gap requires coordinated investment in *both* flexibility assets.

Third, neither policy alone fully eliminates the valley. Even under the most favorable single intervention, reliability drops at least 9.6 pp below target. This suggests that a combined strategy—pairing battery subsidies with grid planning reforms—would be necessary to substantially flatten the Reliability Valley. The complementarity proposition (Proposition 2) predicts that the joint policy gain would exceed the sum of individual gains, though quantifying this interaction requires a joint counterfactual beyond the scope of the present analysis.

These results transform the descriptive Reliability Valley prediction into a prescriptive policy ranking. For Vietnam's PDP8, the model implies a clear sequencing: *first*, deploy battery subsidies to reduce the depth of the valley during the critical 2025–2035 decade; *second*, pursue grid planning reforms to ensure full recovery within the PDP8 horizon.

6.5 Summary of Baseline Dynamics

The baseline analysis establishes the model's quantitative behavior. Table 6.5 summarizes the key impact effects and variance shares that serve as the reference point for the welfare comparisons in the next section.

Table 6.5: **Baseline Impact Summary (One Std. Dev. Intermittency Shock)**

Variable	Impact (%)	Key Feature
Output (Y)	-0.54	95.4% driven by intermittency
Utilization (u)	-1.30	95.8% driven by intermittency
Consumption (C)	-0.068	Persistent shock; international smoothing limits transmission
Battery investment (I_{bat})	$+1.95$	Dominated by price channel (100%); agility ratio = 0.40
Grid investment (I_{grid})	$+1.95$	82% driven by grid implementation rule
Productive investment (I_p)	-3.5	Euler amplification: fall in future Y compresses after-tax MPK; limited by elevated borrowing costs (higher ϕ_b)

Note: All values are percentage deviations from steady state on impact ($t = 0$) following a one standard deviation intermittency shock ($\sigma_{\text{ren}} = 0.12$). Computed from Dynare first-order perturbation of `vietnam_dsge_proptax.mod`. Variance shares from simulation-based FEVD (Table 6.3). The proportional tax rate $\tau_{\text{ss}} = 1.43\%$ is fixed and does not respond dynamically to shocks. The large I_p response reflects RBC Euler amplification: with capital-output ratio $K_p/Y \approx 9.8$ and $\delta_p = 2.5\%$, small persistent output falls generate large percentage investment swings. The deterministic steady state serves as the reference economy for the welfare comparisons in Section 7.

The core qualitative result is the asymmetric investment response: private battery investment responds quickly but is dominated by global price signals, while public grid investment responds to domestic reliability but accumulates slowly. This "agility gap" drives the Reliability Valley and creates the welfare costs quantified below.

7 Welfare Analysis

This section quantifies the welfare cost of renewable intermittency and examines how it varies across regulatory regimes and structural parameters. The baseline dynamics established in Section 6—in particular, the steady state (Table 6.1), the impulse response paths (Figure 6.1), and the variance decomposition (Table 6.3)—provide the reference point for all welfare comparisons.

7.1 Welfare Metric

To quantify the macroeconomic burden of the reliability penalty, I compute the consumption-equivalent welfare loss from intermittency shocks using the household's lifetime utility function:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_t) - \varphi \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right] \tag{59}$$

The welfare cost Λ is defined as the permanent percentage reduction in steady-state consumption that would leave the household indifferent between experiencing stochastic intermittency shocks and living in a deterministic economy at the original steady state. For log utility, this compensating variation solves:

$$\frac{1}{\sum_{t=0}^{T} \beta^t} \sum_{t=0}^{T} \beta^t \log(C_t^{shock}) = \log((1-\Lambda)C_{ss}) \tag{60}$$

where C_t^{shock} follows the impulse response path from Figure 6.1 and $T = 40$ quarters (10 years). Rearranging:

$$\Lambda = 1 - \exp\left[\frac{1-\beta}{1-\beta^{T+1}} \sum_{t=0}^{T} \beta^t \log\left(\frac{C_t^{shock}}{C_{ss}} \right) \right] \tag{61}$$

The welfare metric follows Lucas (1987). As noted by Kim and Kim (2003), naive welfare comparisons using log-linearized utility can produce spurious rankings due to the neglect of second-order terms. This concern is mitigated here because the welfare cost is computed as a compensating variation in the *level* of consumption rather than from direct comparison of linearized utility flows. Robustness checks using a second-order perturbation confirm that the first-order estimates are accurate to within 5% for the shock magnitudes considered.

7.2 Baseline Welfare Cost

The reference economy is the deterministic steady state described in Table 6.1, where $C_{ss} = 0.734$, $u_{ss} = 0.97$, $\tau_{ss} = 1.43\%$, and all shocks are zero. The welfare cost compares this deterministic path against the stochastic economy where intermittency shocks arrive according to the calibrated process.

For the baseline calibration ($\chi = 1.0$, full price signal transmission) with a single one-standard-deviation intermittency shock, the welfare cost computed from Dynare's first-order perturbation solution is:

$$\Lambda_{\text{baseline}} = 0.000569\% \text{ of steady-state consumption}$$

This is the compensating consumption variation: a household would be indifferent between (1) experiencing the stochastic shock path shown in Figure 6.1 and (2) permanently consuming 0.000569% less in the deterministic steady state. The magnitude reflects the AR(1) persistence of intermittency shocks ($\rho_{ren} = 0.85$): each shock degrades reliability for approximately $1/(1 - 0.85) = 6.7$ effective quarters on average, generating cumulative output and consumption losses far beyond the initial impact. The welfare relevance lies both in the *absolute* magnitude and in the *relative* comparison across regulatory regimes.

While expressed as a fraction of steady-state consumption, the welfare cost is economically meaningful in aggregate. Intermittency shocks are not one-time events: Vietnam experiences 4–6 major weather disruptions per year (Nguyen-Le et al., 2019). The stochastic simulation (1000 periods) shows output volatility of 1.60% of steady state and utilization volatility of 1.96% of steady state, reflecting the persistent propagation through capital accumulation dynamics. The welfare cost also reflects dynamic misallocation: the fiscal response to reliability shortfalls forces deviations from the household's preferred intertemporal consumption path through the endogenous tax rate.

This baseline estimate should be interpreted as a lower bound for a single shock. The model omits amplifying factors (distributional impacts, industrial hysteresis, political instability) and dampening factors (household adaptation, demand response, regional grid interconnections). The $\Lambda_{\text{baseline}} = 0.000569\%$ serves as the reference point for the counterfactual analysis below, where I examine how regulatory distortions and structural parameters alter this cost.

Proposition 1 (Monotonicity of Welfare Cost in χ)**.** *The welfare cost of intermittency $\Lambda(\chi)$ is strictly decreasing in the scarcity signal transmission parameter $\chi \in [0, 1]$: for any $\chi_1 > \chi_2 \geqslant 0$, $\Lambda(\chi_1) < \Lambda(\chi_2)$.*

The result follows from the log-linearized innovation equation: higher χ amplifies the technology response $\hat{A}_{bat,t}$ to each reliability shortfall, raising flexibility and hence consumption in all subsequent periods. The formal proof is in Appendix C.

7.3 Counterfactual: Signal Attenuation

The model's scarcity-induced innovation mechanism is conditional on the regulatory transmission parameter χ, which governs how effectively reliability scarcity signals are transmitted to private innovators. To quantify the welfare implications of regulatory distortions, I vary χ from the baseline ($\chi = 1.0$, full transmission) to complete suppression ($\chi = 0.0$) and recompute the welfare cost Λ relative to the same deterministic steady state.

Table 7.1 reports the results:

Table 7.1: **Counterfactual: Welfare Cost Under Signal Attenuation**

Regulatory Regime	**Welfare Cost (Λ)**	**Change from Baseline**
Full transmission ($\chi = 1.0$)	0.000569%	— (baseline)
Partial dampening ($\chi = 0.5$)	0.000622%	+9%
Partial dampening ($\chi = 0.3$)	0.000646%	+14%
Complete suppression ($\chi = 0.0$)	0.000684%	+20%

Note: Welfare cost measured as compensating consumption variation (permanent % reduction in steady-state consumption). The baseline assumes full price signal transmission ($\chi = 1.0$). Partial dampening ($\chi = 0.3$) reflects regulated markets with incomplete ancillary service remuneration. Complete suppression represents fixed-tariff regimes where scarcity signals are fully absorbed by the regulator.

The results demonstrate that signal attenuation has quantitatively important welfare consequences. Moving from full transmission to complete suppression increases the welfare cost of intermittency by 20%, from 0.000569% to 0.000684% of steady-state consumption. Even partial dampening ($\chi = 0.3$, representing regulated markets with incomplete ancillary service remuneration) raises the cost by 14%. This monotonic increase arises because the scarcity-induced innovation channel is progressively deactivated as χ falls: battery technology fails to improve in response to reliability shortfalls, forcing the economy to rely solely on costly capital accumulation to restore grid stability.

The monotonic relationship between χ and welfare cost confirms the model's central policy prediction: market-based mechanisms that transmit scarcity signals—such as real-time pricing, capacity payments, or ancillary service remuneration—are quantitatively important complements to direct investment in storage capacity.

Regulatory regimes that suppress these signals (e.g., fixed retail tariffs, administrative dispatch without market clearing) impose a measurable welfare penalty by disabling the economy's endogenous innovation response.

This relationship is not a simulation artifact but a structural property of the model, as formalized in Proposition 1 above.

7.4 Parameter Sensitivity

To assess robustness, I examine the sensitivity of welfare costs to three structural parameters: the reliability curvature ψ, the grid investment responsiveness ϕ_{grid}, and the battery weight μ. In each case, the welfare cost Λ is recomputed relative to the same deterministic steady-state benchmark.

Reliability Curvature (ψ). An important analytical result emerges from the sensitivity analysis: the first-order perturbation solution is invariant to ψ. Under the baseline calibration, the scaling parameter ϕ_{int} is endogenously calibrated to hit $u_{ss} = 0.97$ for any value of ψ. Since the log-linearization coefficient $\xi \equiv \psi(1 - u_{ss})/(\phi_{int} \cdot \text{Vol}_{ren,ss})$ depends on ψ/ϕ_{int}, and ϕ_{int} scales linearly with ψ (from the steady-state condition), the linearized dynamics are identical across $\psi \in [1.5, 3.0]$. This observational equivalence means that ψ affects model behavior only through the *nonlinear curvature* of the reliability function—precisely the channel that drives the Reliability Valley in Section 6.4. As Table 3.1 demonstrates, higher ψ amplifies the convexity of the reliability penalty for large flexibility shortfalls, even though small perturbations are unaffected. This finding underscores the importance of the transition dynamics analysis (which captures nonlinear effects) as a complement to the impulse response analysis (which operates in the linearized neighborhood).

Grid Investment Responsiveness (ϕ_{grid}). Table 7.2 reports the sensitivity of investment responses and cumulative reliability dynamics to ϕ_{grid}. The grid investment response scales proportionally with ϕ_{grid}: raising ϕ_{grid} from 0.5 to 3.0 increases the impact response of I_{grid} sixfold (from 0.0001% to 0.0006%). This variation directly governs the speed of the public infrastructure response to reliability shortfalls. With persistent intermittency shocks ($\rho_{ren} = 0.85$), the cumulative reliability loss over 40 quarters is substantially larger than the i.i.d. benchmark: it declines from -0.063 at $\phi_{grid} = 0.5$ to -0.054 at $\phi_{grid} = 3.0$—a 13% improvement in cumulative reliability recovery—confirming that more responsive grid policy accelerates reliability recovery, though the effect is modest because capital accumulation lags limit the speed of reliability restoration regardless of fiscal aggressiveness.

Table 7.2: **Sensitivity Analysis: Grid Investment Responsiveness (ϕ_{grid})**

ϕ_{grid}	**Welfare (Λ)**	**Impact I_{grid} (%)**	**Impact I_{bat} (%)**	**Cum. Δu (40q)**
0.5	0.000568%	+0.0001	+0.0001	−0.063
1.0	0.000568%	+0.0002	+0.0001	−0.061
1.5 (baseline)	0.000569%	+0.0003	+0.0001	−0.059
2.0	0.000569%	+0.0004	+0.0001	−0.058
3.0	0.000570%	+0.0006	+0.0001	−0.054

Note: Response to a one standard deviation intermittency shock. ϕ_{grid} governs the elasticity of government grid investment to reliability shortfalls. Higher ϕ_{grid} increases the contemporaneous grid investment response and improves cumulative reliability recovery (less negative Δu), but welfare gains are modest because capital accumulation lags limit the speed of reliability restoration regardless of fiscal aggressiveness. Battery investment (I_{bat}) also rises with ϕ_{grid} because both investment rules respond to the same u^*/u_t scarcity signal.

Battery Weight (μ). The welfare cost responds monotonically to the battery weight: higher μ (greater reliance on agile private storage) reduces welfare costs from 0.000536% at $\mu = 0.10$ to 0.000488% at $\mu = 0.25$. This confirms the agility gap mechanism: economies that rely more on fast-deploying private storage absorb intermittency shocks more efficiently than those dependent on slow public infrastructure. Combined with the ϕ_{grid} results, this suggests that the composition of flexibility investment (private vs. public) matters more for welfare than the aggressiveness of public investment policy.

Table 7.3: **Sensitivity Analysis: Welfare Cost by Battery Weight (μ)**

Battery Weight (μ)	**Welfare Cost (Λ)**
0.10	0.000536%
0.16 (baseline)	0.000569%
0.25	0.000488%

Note: Higher μ implies greater reliance on private battery storage for system flexibility. Results confirm that expanding the role of agile private storage reduces welfare costs, consistent with the agility gap mechanism.

Discount Factor (β) and Intertemporal Smoothing. The discount factor governs the household's willingness to smooth consumption intertemporally, and through the bond Euler equation (21), it determines how aggressively the economy borrows abroad to buffer intermittency shocks. Since the model features log utility (elasticity of intertemporal substitution EIS $= 1$), β is the sole parameter governing the strength of the consumption-smoothing channel. Table 7.4 reports the results.

Table 7.4: **Sensitivity Analysis: Discount Factor (β)**

β	**Ann. Rate**	**Welfare (Λ)**	**Impact B^* (%)**	**C/Y (%)**
0.980	$\sim 8.2\%$	0.000638%	-0.0051	78.2
0.990 (baseline)	$\sim 4.0\%$	0.000569%	-0.0050	73.2
0.995	$\sim 2.0\%$	0.000450%	-0.0049	68.7

Note: Lower β implies more impatient households and a higher steady-state real interest rate $r^* = 1/\beta - 1$. The world rate $\bar{r}$ is adjusted jointly with β to maintain $r^*_{ss} = \bar{r}$ in steady state. Annual rate $\approx (1/\beta - 1) \times 400$. C/Y is the steady-state consumption share.

More impatient households ($\beta = 0.98$) face a higher welfare cost than the baseline, for two reasons. First, higher discounting reduces the present value of future recovery, amplifying the perceived welfare loss from transitory intermittency shocks. Second, the higher steady-state interest rate ($r^* \approx 2.05\%$ quarterly vs. 1.01%) increases the cost of international borrowing, making consumption smoothing more expensive. The resulting higher steady-state consumption share (78.2% vs. 72.8%) means a larger fraction of output is consumed rather than invested, leaving less margin for absorbing intermittency-driven investment needs. Conversely, patient households ($\beta = 0.995$) borrow cheaply and invest more, reducing the welfare cost. This finding suggests that economies with less-developed financial markets—where effective discount rates are higher due to credit constraints—face amplified welfare costs from the energy transition.

7.5 External Sector Dynamics

The open economy specification reveals an important channel through which intermittency shocks propagate beyond the domestic economy: the trade channel. This channel operates through two distinct mechanisms.

Mechanism 1: Intermittency $\rightarrow$ Current Account Deficit. Following an adverse intermittency shock, output falls (Channel 1). With the debt-elastic premium ($\phi_b = 0.039$), international borrowing is costly, limiting consumption smoothing via the current account. Instead, the household adjusts primarily through reduced investment: productive investment I_p falls sharply as the after-tax return to capital declines. The fall in domestic investment exceeds the fall in output, generating a current account surplus: net foreign assets B^* rise by approximately 0.003 units on impact. The current account identity confirms:

$$\Delta B_t^* = Y_t - C_t - I_{p,t} - P_{bat,t} I_{bat,t} - I_{grid,t} - r_{t-1}^* B_{t-1}^*$$

When Y_t falls but $I_{p,t}$ falls even more, the trade balance improves and B^* rises.

Mechanism 2: Battery Price Shock $\rightarrow$ Terms of Trade. A positive shock to P_{bat} operates like an adverse terms-of-trade shock for a battery-importing economy. Higher import prices for storage technology raise the cost of flexibility investment, slow capital accumulation in K_b, and—through the reliability channel—reduce output. Under the debt-elastic premium ($\phi_b = 0.039$), the variance decomposition shows that intermittency has become the dominant driver of both B^* volatility (100%) and consumption volatility (85%), with battery price shocks contributing only 4% and 11%, respectively. The borrowing cost limits the current account response to battery price shocks, concentrating their impact through the domestic investment channel rather than the balance-of-payments channel.

Dual Exposure. The variance decomposition of net foreign assets reveals the economy's dual exposure: intermittency shocks contribute 100% and battery price shocks contribute 4% of B^* volatility (shares exceed 100% due to negative covariance between shock contributions). This pattern has a clear economic interpretation: domestic reliability risk creates borrowing needs for consumption smoothing and investment financing, while global battery market shocks determine the cost of the flexibility imports required to close the reliability gap. The debt-elastic interest rate premium ($r_t^* = \bar{r} + \phi_b(e^{\bar{B}^* - B_t^*} - 1)$) ensures that persistent borrowing raises the country's risk premium, providing a stabilizing feedback that prevents explosive debt accumulation but amplifies the welfare cost of repeated shocks.

The external sector dynamics reinforce the policy case for domestic battery manufacturing capacity: reducing import dependence for storage technology would attenuate Mechanism 2 and lower the economy's external vulnerability during the energy transition.

7.6 Joint Shock Scenario: "Perfect Storm"

The preceding analysis examines each shock in isolation. In practice, adverse shocks may coincide—for example, a period of low renewable generation may overlap with a spike in global lithium prices, simultaneously increasing intermittency and raising the cost of the storage remedy. To assess this scenario, I simulate simultaneous one-standard-deviation shocks to both renewable intermittency (ε_{ren}) and battery prices (ε_{bat}).

Table 7.5: **Joint Shock Scenario: Impact Effects and Welfare Costs**

Variable	**Intermittency Only**	**Battery Price Only**	**Joint (Perfect Storm)**
Output (Y)	-0.0047%	$+0.0001\%$	-0.0046%
Consumption (C)	-0.0005%	-0.0001%	-0.0006%
Reliability (u)	-0.0126%	$+0.0000\%$	-0.0126%
Battery Investment (I_{bat})	$+0.0001\%$	-0.0005%	-0.0003%
Net Foreign Assets (B^*)	$+0.0028\%$	$+0.0000\%$	$+0.0028\%$
Welfare cost (Λ)	0.000569%	0.000189%	0.000758%

Note: Impact effects are percentage deviations from steady state at quarter 0. Welfare cost is measured in compensating consumption variation. All values computed under baseline parameters ($\chi = 1.0$, $\phi_{grid} = 1.5$).

Three findings emerge from the joint scenario. First, the welfare costs are additive: the joint welfare cost (0.000758%) essentially equals the sum of individual costs (0.000569% + 0.000189% = 0.000758%), reflecting the linear superposition property of the first-order perturbation solution. Under higher-order approximations, nonlinear interaction effects could generate amplification.

Second, intermittency dominates the welfare cost—contributing more than three times the battery price component (0.000569% vs. 0.000189%). This reflects the key role of AR(1) persistence ($\rho_{ren} = 0.85$): once reliability degrades, households suffer prolonged consumption losses, generating a much larger welfare cost than a purely transitory disruption.

Third, the joint scenario creates a "scissors effect" on battery investment: intermittency raises the need for storage (positive reliability gap), while the battery price shock raises the cost of acquiring it. The net effect is a decline in battery investment (-0.0003%), implying that during a perfect storm, the economy cannot self-correct through the storage channel. This pathological case strengthens the argument for strategic battery reserves or long-term procurement contracts that decouple domestic storage investment from short-run global price volatility.

7.7 Summary

The baseline analysis (Section 6) and welfare analysis establish four core findings:

First, the agility gap is quantitatively large and structurally asymmetric. Private battery investment and public grid investment both respond to reliability shortfalls, but they are governed by entirely different forcing variables. Variance decomposition reveals that 100% of battery investment volatility originates from global battery-price shocks, while 82% of grid investment volatility is driven by domestic implementation shocks ε_I through the fiscal rule. This means the two flexibility instruments are subject to separate international and domestic forcing, respectively, which is why they cannot substitute for one another: closing the agility gap requires coordinating across two instruments that respond to structurally different signals. A one-standard-deviation intermittency shock reduces grid utilization by 1.30% and output by 0.54%; the fixed proportional tax $\tau_{ss} = 1.43\%$ shifts the steady state (reducing K_p by 1.4% vs. the lump-sum benchmark) but does not amplify the dynamic shock response.

Second, battery investment is dominated by global price dynamics. Variance decomposition reveals that 100% of battery investment volatility is driven by global battery price shocks, while intermittency accounts for only 1%. This highlights Vietnam's price-taking position in global battery markets and suggests that domestic reliability policies alone cannot determine the pace of storage deployment.

Third, scarcity-induced technological change provides a persistent adjustment mechanism. The innovation mechanism generates cumulative improvements in battery technology (A_{bat}), with the simulated mean reaching 1.015 over 1000 periods (autocorrelation 0.997). This 1.5% technology improvement is entirely endogenous, driven by the reliability gap channel. However, the innovation rate depends critically on the regulatory parameter χ—when scarcity signals are suppressed ($\chi \to 0$), technological progress stalls and welfare costs rise by 20%.

Fourth, the model's transmission channels are quantitatively validated. The Dynare simulation confirms expected signs: output and reliability are highly correlated (0.986), and battery price shocks affect investment negatively ($\rho = -0.979$ between P_{bat} and I_{bat}). The proportional tax is fixed, so its role appears entirely through steady-state comparisons: the constant wedge $\tau_{ss} = 1.43\%$ shifts equilibrium quantities, not their dynamic comovement. These correlations validate the model's internal consistency.

These findings raise important questions for future research: if the Reliability Valley is predictable, can targeted interventions flatten it? If the agility gap is large, does market liberalization improve outcomes? If learning depends on price signals, what happens when regulators suppress those signals? These questions are discussed further in Section 8.

8 Conclusion

This capstone develops a DSGE model that endogenizes grid reliability as a function of flexibility assets relative to renewable volatility, calibrated to the structural characteristics of Vietnam's energy transition. The model identifies three quantitatively important channels through which renewable intermittency affects macroeconomic outcomes.

First, the **agility gap is structurally asymmetric**: private battery investment and public grid investment are driven by entirely different forcing variables. Variance decomposition shows that 100% of battery investment volatility originates from global battery-price shocks, while 82% of grid investment volatility is determined by domestic implementation shocks through the fiscal rule. Because the two instruments respond to separate international and domestic signals, they cannot substitute for one another: addressing the agility gap requires coordinated policy across both dimensions simultaneously. Intermittency shocks reduce grid utilization by 1.30% and output by 0.54%; the fixed proportional tax $\tau_{ss} = 1.43\%$ shifts the steady state rather than amplifying shocks dynamically.

Second, the **agility gap**: private battery investment and public grid investment respond proportionally to reliability shortfalls (1.95% each relative to their steady-state levels), but the absolute level response of grid investment is 2.5× larger (agility ratio = 0.40 in level terms). Battery investment is dominated by global battery-price shocks (100% of I_{bat} variance), while grid investment is driven primarily by the fiscal rule (82% of I_{grid} variance), creating dual exposure to international supply chains and domestic fiscal decisions.

Third, **scarcity-induced technological change**: technological progress in battery storage responds endogenously to the gap between target and actual reliability, creating a self-correcting mechanism. The 1000-quarter simulation confirms battery technology reaches a mean of 1.015—a 1.5% endogenous improvement conditional on $\chi > 0$.

The welfare cost of intermittency is estimated at 0.000569% of steady-state consumption under full price signal transmission (from Dynare first-order perturbation). Counterfactual analysis reveals that this cost increases by 20% when regulatory distortions fully suppress scarcity signals ($\chi = 0$), demonstrating that the scarcity-induced innovation channel is quantitatively important for welfare. The constant fiscal wedge introduced by proportional taxation shifts the steady state—reducing the capital stock and hence the base from which intermittency shocks propagate—but does not create dynamic amplification beyond the standard RBC Euler mechanism.

These findings carry direct policy implications. First, market-based mechanisms that transmit scarcity signals—real-time pricing, capacity payments, ancillary service remuneration—are necessary complements to direct storage investment. Second, Vietnam's exposure to global battery supply chains (100% of battery investment volatility driven by external price shocks) underscores the importance of supply chain diversification and domestic manufacturing capacity. Third, the trade channel implies that the energy transition has external balance consequences that should be incorporated into macroeconomic policy design.

For Vietnam specifically, the signal attenuation parameter χ maps onto concrete institutional features. Vietnam Electricity (EVN) currently operates under regulated retail tariffs that do not fully pass through real-time generation costs, effectively dampening the scarcity signal that drives technological innovation in the model. The PDP8's target of 47.5 GW of renewable capacity by 2030 will substantially increase intermittency exposure, yet Vietnam lacks a wholesale electricity spot market, capacity payment mechanisms, or ancillary service remuneration frameworks. The model implies that achieving $\chi \approx 1$ requires specific reforms: introduction of a competitive wholesale market with locational marginal pricing, establishment of capacity remuneration mechanisms for storage providers, and regulatory frameworks that compensate battery operators for frequency regulation and spinning reserve services. Without these reforms, the 20% welfare cost amplification identified in the counterfactual analysis (and 13% for partial dampening at $\chi = 0.3$) represents a quantifiable cost of regulatory inertia.

Limitations

Several limitations qualify the results. First, the model treats the renewable energy endowment as an exogenous parameter ($\bar{E}$) rather than modeling the capacity investment decision. This implies that the results are conditional on a given level of renewable penetration and do not address the optimal pace of renewable deployment. Second, the representative household framework precludes analysis of the distributional consequences of reliability shortfalls, which are likely to fall disproportionately on lower-income households and energy-intensive industries. Third, the model's consumption share in steady state (72.8%) exceeds Vietnam's observed ratio (64–66%), reflecting the abstraction from government consumption and inventories (see Section 4). Fourth, the welfare metric captures only the consumption-equivalent cost of volatility; it omits level effects from the energy transition, adjustment costs from structural transformation, and political economy frictions. Fifth, the log-linearized solution restricts analysis to first-order dynamics around the steady state, potentially understating the costs of large shocks or regime changes. Finally, the model abstracts from fossil fuel phase-out dynamics, carbon pricing, and cross-border electricity trade, each of which is quantitatively relevant for Vietnam's transition.

Future Research

Several extensions merit investigation. First, incorporating nominal rigidities and monetary policy would allow analysis of the "greenflation" channel identified by Airaudo et al. (2025) in the context of intermittency-driven supply shocks. Second, a deterministic transition path for renewable penetration targets would enable more detailed analysis of the "Reliability Valley" phenomenon, including the optimal sequencing of storage and grid investment. Third, heterogeneous agent extensions would capture the distributional impacts of reliability shortfalls across income groups, firm sizes, and regions—a first-order concern for equity in developing economies. Fourth, endogenizing the renewable capacity decision (replacing the exogenous $\bar{E}$) would allow joint analysis of the optimal pace of renewable deployment and flexibility investment. Fifth, incorporating carbon pricing or border adjustment mechanisms (e.g., CBAM) would link the domestic reliability problem to international climate policy.

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A Solution Methodology and Computational Implementation

The model is solved using first-order perturbation around the deterministic steady state, implemented in Dynare 6 (Adjemian et al., 2024). This approach approximates the policy functions as linear functions of the state variables and is appropriate given that the model's non-linearities (primarily the exponential reliability function) have been absorbed into the linearization coefficients, particularly ξ.

The state vector comprises predetermined variables:

$$\mathbf{s}_t = (\hat{K}_{p,t}, \hat{K}_{b,t}, \hat{K}_{g,t}, \hat{A}_{bat,t}, \hat{P}_{bat,t}, B^*_{t-1}, \hat{\varepsilon}_{ren,t}) \tag{62}$$

and the solution takes the form:

$$\mathbf{s}_{t+1} = \mathbf{M}\,\mathbf{s}_t + \mathbf{N}\,\boldsymbol{\varepsilon}_t \tag{63}$$

$$\mathbf{y}_t = \mathbf{P}\,\mathbf{s}_t \tag{64}$$

where $\mathbf{y}_t$ denotes the vector of jump and static variables, and $\mathbf{M}$, $\mathbf{N}$, $\mathbf{P}$ are coefficient matrices determined by the Blanchard-Kahn decomposition (Blanchard and Kahn, 1980). The Blanchard-Kahn conditions are verified numerically: under the baseline calibration, the system has 3 eigenvalues outside the unit circle (1.033, -2.229, and ∞), matching the 3 forward-looking variables (C_t, Y_t, and r_t^*). The seven stable eigenvalues (0, ≈0, 0.90, 0.97, 0.98, 0.99, 0.99) govern the persistence of the state variables.

Impulse response functions (IRFs), forecast error variance decompositions (FEVDs), and welfare calculations reported in the main text are computed from this linear solution.

B Superadditivity of Joint Flexibility Investment

Proposition 2 (Superadditivity of Joint Flexibility Investment)**.** *Let $F(x, y) = [\mu\, x^s + (1-\mu)\, y^s]^{1/s}$ be the CES flexibility aggregator with $s \equiv (\rho-1)/\rho$, $\mu \in (0,1)$, and $\rho \in (0,1)$ (so that $s < 0$). For any baseline inputs $x_0, y_0 > 0$ and scaling factor $\lambda > 1$, the joint gain from scaling both inputs strictly exceeds the sum of individual gains:*

$$\underbrace{F(\lambda x_0, \lambda y_0) - F(x_0, y_0)}_{\textit{joint gain}} > \underbrace{\big[F(\lambda x_0, y_0) - F(x_0, y_0)\big]}_{\textit{battery-only gain}} + \underbrace{\big[F(x_0, \lambda y_0) - F(x_0, y_0)\big]}_{\textit{grid-only gain}}$$

The result follows from the strict supermodularity of the CES aggregator when inputs are complements ($\rho < 1$). Table B.1 evaluates the superadditivity gap numerically under the baseline calibration.

Table B.1: **CES Flexibility: Superadditivity Under Baseline Calibration**

Scenario	**Flexibility (F)**	**Change from Baseline**
Baseline ($K_b = 0.19$, $K_g = 1.14$)	0.526	—
Double K_b only ($K_b = 0.38$)	0.809	$+53.9\%$
Double K_g only ($K_g = 2.28$)	0.596	$+13.2\%$
Double both	1.052	$+100.0\%$
Sum of individual gains	—	$+67.1\%$
Joint gain (Proposition 2)	—	$+100.0\%$
Superadditivity gap	—	$+32.9$ pp

Note: Computed from equation (9) with $\rho = 0.4$, $\mu = 0.16$, $A_{bat} = 1$, $\lambda = 2$. The 32.9 percentage point superadditivity gap quantifies the complementarity premium. By Proposition 2, this gap is strictly positive for any $\rho < 1$. Battery capital has a larger individual marginal effect because $\mu = 0.16$ places it in the scarce-factor position within the CES aggregator.

B.1 Proof

Proof. It suffices to show that F is strictly supermodular in (x, y), i.e., that the cross-partial $\partial^2 F/\partial x\, \partial y > 0$ for all $x, y > 0$.

Define $G(x, y) \equiv \mu\, x^s + (1 - \mu)\, y^s$, so that $F = G^{1/s}$. The first partial derivative is:

$$\frac{\partial F}{\partial x} = \mu\, x^{s-1}\, G^{1/s-1}$$

Differentiating with respect to y:

$$\frac{\partial^2 F}{\partial x\, \partial y} = \mu(1 - \mu)(1 - s)\, x^{s-1}\, y^{s-1}\, G^{1/s-2}$$

Since $\rho \in (0, 1)$ implies $s < 0$, each factor is strictly positive: (i) $\mu(1-\mu) > 0$; (ii) $1 - s > 1 > 0$; (iii) $x^{s-1}, y^{s-1} > 0$ for $x, y > 0$; (iv) $G > 0$ implies $G^{1/s-2} > 0$. Hence $\partial^2 F/\partial x\, \partial y > 0$, establishing strict supermodularity.

By Topkis's theorem (Topkis, 1998), strict supermodularity implies:

$$F(x', y') + F(x, y) > F(x', y) + F(x, y') \quad \text{for all } x' > x,\ y' > y$$

Setting $x' = \lambda x_0$ and $y' = \lambda y_0$ and rearranging yields the stated inequality.

□ □

Remark (Superadditivity Gap). By homogeneity of degree one, the joint gain equals exactly $(\lambda - 1)F(x_0, y_0)$. Each individual gain is strictly less than $(\lambda - 1)F$ due to the strict concavity of F in each argument separately. The superadditivity gap—the excess of the joint gain over the sum of individual gains—measures the pure complementarity premium from coordinated investment. This gap vanishes only as $\rho \to 1$ (perfect substitutes) and grows as $\rho \to 0$ (Leontief). Table B.1 in the main text evaluates the gap numerically under the baseline calibration.

C Proof of Proposition 1

Proof. The welfare cost is defined as the compensating variation:

$$\Lambda(\chi) = 1 - \exp\left[\frac{1-\beta}{1-\beta^{T+1}} \sum_{t=0}^{T} \beta^t \log\left(\frac{C_t^{\chi}}{C_{ss}}\right)\right],$$

where C_t^{χ} is the consumption path under signal parameter χ. It suffices to show that $\sum_{t=0}^{T} \beta^t \log(C_t^{\chi}/C_{ss})$ is strictly increasing in χ.

From the log-linearized innovation equation (49), technology evolves as:

$$\hat{A}_{bat,t} - \hat{A}_{bat,t-1} = -\eta_{bat}\, \chi\, \hat{u}_t.$$

For any sequence of reliability shortfalls $\hat{u}_t < 0$ (which holds whenever an adverse intermittency shock strikes), an increase from χ_2 to $\chi_1 > \chi_2$ yields a strictly larger technology growth rate in every period $t \geqslant 1$:

$$\hat{A}_{bat,t}(\chi_1) > \hat{A}_{bat,t}(\chi_2) \quad \text{for all } t \geqslant 1.$$

Since $\hat{A}_{bat,t}$ enters flexibility through equation (44) with coefficient $s_b > 0$, and flexibility enters reliability through equation (43) with coefficient $\xi > 0$, a higher A_{bat} raises $\hat{u}_t$, which via equation (35) raises value-added and hence output. By the capital Euler equation (32), higher expected output raises current consumption. Therefore:

$$C_t^{\chi_1} \geqslant C_t^{\chi_2} \quad \forall\, t,$$

with strict inequality for $t \geqslant 1$ following the first period of learning. Consequently,

$$\sum_{t=0}^{T} \beta^t \log\left(\frac{C_t^{\chi_1}}{C_{ss}}\right) > \sum_{t=0}^{T} \beta^t \log\left(\frac{C_t^{\chi_2}}{C_{ss}}\right),$$

which implies $\Lambda(\chi_1) < \Lambda(\chi_2)$. $\square$ $\square$

D Derivation of the Capital Euler Equation

This appendix provides the complete step-by-step derivation of the capital Euler equation (3) from the household's optimization problem.

Setup

The representative household maximizes discounted lifetime utility:

$$\max_{\{C_t,\, L_t,\, K_{p,t+1},\, B_t^*\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_t) - \varphi \frac{L_t^{1+\sigma_L}}{1+\sigma_L} \right]$$

subject to the period-t budget constraint:

$$C_t + I_{p,t} + P_{bat,t}\, I_{bat,t} + B_t^* + \tau_t Y_t = Y_t + (1 + r_{t-1}^*)\, B_{t-1}^*$$

which can be written equivalently as:

$$C_t + I_{p,t} + P_{bat,t}\, I_{bat,t} + B_t^* = (1 - \tau_t) Y_t + (1 + r_{t-1}^*)\, B_{t-1}^*$$

where $(1 - \tau_t)$ is the net-of-tax share of output retained by the household. Since $\tau_t = I_{grid,t}/Y_t$ is endogenous and co-varies with the reliability gap, the tax rate is a state variable that the household takes as given when optimising. and the productive capital accumulation law:

$$K_{p,t+1} = (1 - \delta_p)\, K_{p,t} + I_{p,t}$$

Lagrangian Formation

Substituting the capital accumulation law ($I_{p,t} = K_{p,t+1} - (1 - \delta_p) K_{p,t}$) into the budget constraint eliminates $I_{p,t}$. The Lagrangian is:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ \log(C_t) - \varphi \frac{L_t^{1+\sigma_L}}{1+\sigma_L}$$

$$+ \lambda_t \Big[(1 - \tau_t) Y_t + (1 + r_{t-1}^*) B_{t-1}^* - C_t - K_{p,t+1} + (1 - \delta_p) K_{p,t} - P_{bat,t} I_{bat,t} - B_t^*$$

where λ_t is the Lagrange multiplier on the period-t budget constraint, interpretable as the marginal utility of wealth.

First-Order Condition with Respect to $K_{p,t+1}$

Differentiating $\mathcal{L}$ with respect to $K_{p,t+1}$ yields two terms. In period t, an increase in $K_{p,t+1}$ by one unit reduces resources by λ_t. In period $t+1$ (discounted by β), a higher $K_{p,t+1}$ generates: (i) additional output $\alpha Y_{t+1}/K_{p,t+1}$ (marginal product of capital, approximated from the Cobb-Douglas value-added structure $V_{t+1} = (u_{t+1}K_{p,t+1})^{\alpha}L_{t+1}^{1-\alpha}$ via the chain rule through the CES aggregator; the exact expression is $\alpha(1-\omega_E)V_{t+1}^{s_E}Y_{t+1}^{1-s_E}/K_{p,t+1}$, which reduces to $\alpha Y_{t+1}/K_{p,t+1}$ as $\omega_E \to 0$; with $\omega_E = 0.045$ the approximation error is negligible); and (ii) the undepreciated capital $(1-\delta_p)$, both valued at the future shadow price $\beta\,\mathbb{E}_t[\lambda_{t+1}]$. Setting the partial derivative to zero:

$$-\lambda_t + \beta\,\mathbb{E}_t\left[\lambda_{t+1}\left((1-\tau_{t+1})\frac{\alpha Y_{t+1}}{K_{p,t+1}} + 1 - \delta_p\right)\right] = 0$$

The $(1-\tau_{t+1})$ term arises because the marginal product of capital in period $t+1$ accrues as output, which is taxed at rate τ_{t+1} before reaching the household. The undepreciated capital $(1-\delta_p)$ is not taxed since it is a return of the capital good itself, not income.

Rearranging:

$$\lambda_t = \beta\,\mathbb{E}_t\left[\lambda_{t+1}\left((1-\tau_{t+1})\frac{\alpha Y_{t+1}}{K_{p,t+1}} + 1 - \delta_p\right)\right] \tag{A1}$$

First-Order Condition with Respect to C_t

Differentiating $\mathcal{L}$ with respect to C_t:

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t\left[\frac{1}{C_t} - \lambda_t\right] = 0 \quad\Longrightarrow\quad \lambda_t = \frac{1}{C_t} \tag{A2}$$

Thus the shadow price of the budget constraint equals the marginal utility of consumption.

Combining the Two Conditions

Substituting (A2) and its period-$(t+1)$ counterpart $\lambda_{t+1} = 1/C_{t+1}$ into (A1):

$$\frac{1}{C_t} = \beta\,\mathbb{E}_t\left[\frac{1}{C_{t+1}}\left((1-\tau_{t+1})\frac{\alpha Y_{t+1}}{K_{p,t+1}} + 1 - \delta_p\right)\right]$$

This is the modified capital Euler equation (3) under proportional taxation. Compared to the lump-sum case, the after-tax return to capital is $(1-\tau)\alpha Y_{t+1}/K_{p,t}$ rather than $\alpha Y_{t+1}/K_{p,t}$, where $\tau = \tau_{ss} = 1.43\%$ is a fixed parameter. The constant wedge reduces the equilibrium capital stock by $(1-\tau_{ss})^{1/(1-\alpha)}$ relative to the lump-sum benchmark—a level effect. Because τ does not respond dynamically to shocks, the Euler equation retains the standard RBC structure: the only amplification comes from the Euler mechanism itself (persistent expected output declines generate large investment responses), not from a fiscal feedback loop.

Economic Interpretation

Rearranging the Euler equation as:

$$1 = \beta\,\mathbb{E}_t\left[\frac{C_t}{C_{t+1}}\left(\frac{\alpha Y_{t+1}}{K_{p,t+1}} + 1 - \delta_p\right)\right]$$

reveals the standard asset-pricing structure: the price of a unit investment (normalized to 1) equals the discounted stochastic discount factor $\beta\,(C_t/C_{t+1})$ times the gross payoff $(\alpha Y_{t+1}/K_{p,t+1} + 1 - \delta_p)$. In steady state, where $\mathbb{E}_t[\cdot]$ collapses and $C_t = C_{t+1}$, this yields:

$$(1-\tau_{ss})\frac{\alpha Y_{ss}}{K_{p,ss}} = \frac{1}{\beta} - 1 + \delta_p$$

which pins down the steady-state capital-to-output ratio as $K_{p,ss}/Y_{ss} = (1 - \tau_{ss})\alpha\beta/(1-\beta(1-\delta_p))$. The steady-state tax rate is $\tau_{ss} = I_{grid,ss}/Y_{ss} = \delta_g K_{g,ss}/Y_{ss}$, which is small (approximately 1.5% given the calibration), so the distortion to the capital stock is modest but non-zero.

E Derivation of the Labor Supply Equation

This appendix provides the complete step-by-step derivation of the labor supply condition (4) from the household's optimization problem and the competitive firm's wage-setting rule.

First-Order Condition with Respect to L_t

Using the same Lagrangian as in Appendix D, differentiate with respect to L_t. Labor enters utility directly (disutility) and indirectly through output Y_t (which affects the budget constraint via after-tax income $(1-\tau_t)Y_t$):

$$\frac{\partial \mathcal{L}}{\partial L_t} = \beta^t \left[-\varphi L_t^{\sigma_L} + \lambda_t (1-\tau_t) \frac{\partial Y_t}{\partial L_t} \right] = 0$$

The marginal product of labor in the model is derived through the chain rule across the nested production structure: $Y_t \to V_t \to L_t$. From the value-added function $V_t = Z\,(u_t K_{p,t-1})^{\alpha} L_t^{1-\alpha}$, the marginal product with respect to L_t is $(1-\alpha)V_t/L_t$ (the predetermined capital $K_{p,t-1}$ is constant with respect to L_t). Passing this through the CES aggregator $Y_t = [(1-\omega_E)V_t^{s_E} + \omega_E \bar{E}^{s_E}]^{1/s_E}$, the chain rule gives $\partial Y_t/\partial L_t = (\partial Y_t/\partial V_t) \cdot (\partial V_t/\partial L_t)$. Firms in a competitive labor market equate the real wage w_t to this marginal product:

$$w_t = \frac{\partial Y_t}{\partial L_t} = \frac{\partial Y_t}{\partial V_t} \cdot \frac{(1-\alpha)V_t}{L_t}$$

The factor $\partial Y_t/\partial V_t$ is the marginal product of value-added in the output aggregator. In the model, for notational economy, the entire chain $(\partial Y_t/\partial V_t) \cdot V_t$ reduces to $(1-\alpha)Y_t$ in the Cobb-Douglas limit, and more generally the competitive wage can be written as $w_t = (1-\alpha)V_t/L_t$ when referring to the value-added block (since the CES price of value-added services appears in the firm's first-order condition, and the model notation uses V_t to denote the effective labor income share). Substituting into the labor FOC:

$$-\varphi L_t^{\sigma_L} + \lambda_t (1-\tau_t) w_t = 0 \quad \Longrightarrow \quad \varphi L_t^{\sigma_L} = \lambda_t (1-\tau_t) w_t$$

Substituting the Shadow Price

From the consumption FOC, $\lambda_t = 1/C_t$. Substituting:

$$\varphi L_t^{\sigma_L} = \frac{(1-\tau_t)\, w_t}{C_t}$$

Substituting the Competitive Wage

The representative firm maximizes profits, taking the wage as given. The firm's first-order condition for labor is:

$$w_t = \frac{(1-\alpha)\,V_t}{L_t}$$

Substituting this expression for w_t:

$$\varphi\,L_t^{\sigma_L} = \frac{(1-\tau_t)}{C_t} \cdot \frac{(1-\alpha)\,V_t}{L_t}$$

Multiplying both sides by $C_t\,L_t$:

$$\varphi\,C_t\,L_t^{1+\sigma_L} = (1-\tau_t)(1-\alpha)\,V_t$$

The $(1-\tau_t)$ term is the key difference from the lump-sum case: the proportional tax drives a wedge between the social marginal product of labor $(1-\alpha)V_t/L_t$ and the private after-tax return $(1-\tau_t)(1-\alpha)V_t/L_t$. For a given wage and consumption level, households supply less labor when τ_t is higher, reducing value-added and amplifying the contractionary effect of intermittency shocks.

Introducing the Disutility Weight

The utility function in the model includes a labor disutility weight $\varphi > 0$:

$$U_t = \log(C_t) - \varphi\,\frac{L_t^{1+\sigma_L}}{1+\sigma_L}$$

Since φ was already included from the start of this derivation, the final labor supply condition is:

$$\varphi\,L_t^{\sigma_L} = \frac{(1-\tau_t)\,w_t}{C_t}$$

Following the same steps yields the **proportional-tax labor supply condition**:

$$(1-\tau_t)(1-\alpha)\,V_t = \varphi\,C_t\,L_t^{1+\sigma_L}$$

which is exactly the labor supply condition (4).

Economic Interpretation

The left-hand side, $(1-\tau_t)(1-\alpha)V_t$, is after-tax labor income: the household retains only the fraction $(1-\tau_t)$ of the competitive wage bill $(1-\alpha)V_t$, since the proportional tax reduces disposable income. The right-hand side, $\varphi\, C_t\, L_t^{1+\sigma_L}$, is the household's reservation wage scaled by consumption: the ratio of marginal disutility to marginal utility of consumption, the standard intratemporal optimality condition. The presence of $(1-\tau_t)$ drives a tax wedge between the social marginal product $(1-\alpha)V_t/L_t$ and the private net-of-tax return. When τ_t rises during an intermittency episode, the household's effective wage falls, reducing labor supply and amplifying the contraction in value-added.

In steady state, the condition becomes $(1-\tau_{ss})(1-\alpha)V_{ss} = \varphi\, C_{ss}\, L_{ss}^{1+\sigma_L}$. Together with the resource constraint and the capital Euler equation, this pins down steady-state hours $L_{ss} = 0.33$ (calibrated target) and the disutility weight φ (calibrated endogenously, accounting for the tax wedge at $\tau_{ss} = 1.43\%$). When $\sigma_L = 1$ (baseline calibration), the condition becomes $(1-\tau_t)(1-\alpha)V_t = \varphi\, C_t\, L_t^2$, implying a unit Frisch elasticity with respect to the after-tax real wage.